Chapter V
Local Class Field Theory

§ 1. The Local Reciprocity Law

The abstract class field theory that we have developed in the last chapter is now going to be applied to the case of a *local field*, i.e., to a field which is complete with respect to a discrete valuation, and which has a finite residue class field. By chap. II, (5.2), these are precisely the finite extensions K of the fields $\mathbb{Q}_p$ or $\mathbb{F}_p((t))$. We will use the following notation. Let

v_K be the discrete valuation normalized by $v_K(K^*) = \mathbb{Z}$,

$\mathcal{O}_K \;= \{a \in K \mid v_K(a) \geq 0\}$ the valuation ring,

$\mathfrak{p}_K \;= \{a \in K \mid v_K(a) > 0\}$ the maximal ideal,

$\kappa \;= \mathcal{O}_K/\mathfrak{p}_K$ the residue class field,

$U_K \;= \{a \in K^* \mid v_K(a) = 0\}$ the unit group,

$U_K^{(n)} \;= 1 + \mathfrak{p}_K^n$ the group of n-th higher units, $n = 1, 2, \ldots,$

$q \;= q_K = \#\kappa,$

$|a|_\mathfrak{p} \;= q^{-v_K(a)}$ the normalized $\mathfrak{p}$-adic absolute value,

μ_n the group of n-th roots of unity, and $\mu_n(K) = \mu_n \cap K^*$.

π_K, or simply π, denotes a prime element of K, i.e., $\mathfrak{p}_K = \pi\mathcal{O}_K$.

In local class field theory, the rôle of the profinite group G of abstract class field theory is taken by the absolute Galois group $G(\bar{k}|k)$ of a fixed local field k, and that of the G-module A by the multiplicative group $\bar{k}^*$ of the separable closure $\bar{k}$ of k. For a finite extension $K|k$ we thus have $A_K = K^*$, and the crucial point is to verify for the multiplicative group of a local field the class field axiom:

(1.1) Theorem. *For a cyclic extension $L|K$ of local fields, one has*

$$\#H^i\big(G(L|K), L^*\big) = \begin{cases} [L:K] & \textit{for } i = 0, \\ 1 & \textit{for } i = -1. \end{cases}$$

Proof: For $i = -1$ this is the claim of proposition (3.5) ("Hilbert 90") in chap. IV. So all we have to show is that the Herbrand quotient is $h(G, L^*) = \#H^0(G, L^*) = [L : K]$, where we have put $G = G(L|K)$. The exact sequence

$$1 \longrightarrow U_L \longrightarrow L^* \xrightarrow{v_L} \mathbb{Z} \longrightarrow 0,$$

in which $\mathbb{Z}$ has to be viewed as the trivial G-module, yields, by chap. IV, (7.3),

$$h(G, L^*) = h(G, \mathbb{Z})\, h(G, U_L) = [L : K]\, h(G, U_L).$$

Hence we have to show that $h(G, U_L) = 1$. For this we choose a normal basis $\{\alpha^\sigma \mid \sigma \in G\}$ of $L|K$ (see [93], chap. VIII, § 12, th. 20), $\alpha \in \mathcal{O}_L$, and consider in $\mathcal{O}_L$ the open (and closed) G-module $M = \sum_{\sigma \in G} \mathcal{O}_K \alpha^\sigma$. Then the open sets

$$V^n = 1 + \pi_K^n M, \quad n = 1, 2, \ldots,$$

form a basis of open neighbourhoods of 1 in U_L. Since M is open, we have $\pi_K^N \mathcal{O}_L \subseteq M$ for suitable N, and for $n \geq N$ the V^n are even subgroups (of finite index) of U_L, because we have

$$(\pi_K^n M)(\pi_K^n M) = \pi_K^{2n} MM \subseteq \pi_K^{2n} \mathcal{O}_L \subseteq \pi_K^{2n-N} M \subseteq \pi_K^n M.$$

Hence $V^n V^n \subseteq V^n$, and since $1 - \pi_K^n \mu$, for $\mu \in M$, lies in V^n, so does $(1 - \pi_K^n \mu)^{-1} = 1 + \pi_K^n (\sum_{i=1}^{\infty} \mu^i \pi_K^{n(i-1)})$. Via the correspondence $1 + \pi_K^n \alpha \mapsto \alpha \bmod \pi_K M$, we obtain G-isomorphisms as in II, (3.10),

$$V^n / V^{n+1} \cong M / \pi_K M = \bigoplus_{\sigma \in G} (\mathcal{O}_K / \mathfrak{p}_K) \alpha^\sigma = \operatorname{Ind}_G(\mathcal{O}_K / \mathfrak{p}_K).$$

So by chap. IV, (7.4), we have $H^i(G, V^n / V^{n+1}) = 1$ for $i = 0, -1$ and $n \geq N$. This in turn implies that $H^i(G, V^n) = 1$ for $i = 0, -1$ and $n \geq N$. Indeed, if for instance $i = 0$ and $a \in (V^n)^G$, then $a = (N_G b_0) a_1$, with $b_0 \in V^n$, $a_1 \in (V^{n+1})^G$, and thus $a_1 = (N_G b_1) a_2$, for some $b_1 \in V^{n+1}$, $a_2 \in (V^{n+2})^G$, etc.; in general,

$$a_i = (N_G b_i) a_{i+1}, \quad b_i \in V^{n+i}, \quad a_{i+1} \in (V^{n+i+1})^G.$$

This yields $a = N_G b$, with the convergent product $b = \prod_{i=0}^{\infty} b_i \in V^n$, so that $H^0(G, V^n) = 1$. In the same way we have for $a \in V^n$ such that $N_G a = 1$, that $a = b^{\sigma - 1}$, for some $b \in V^n$, where σ is a generator of G. Thus $H^{-1}(G, V^n) = 1$. We now obtain

$$h(G, U_L) = h(G, U_L / V^n)\, h(G, V^n) = 1$$

because U_L / V^n is finite. □

(1.2) Corollary. *If $L|K$ is an unramified extension of local fields, then for $i = 0, -1$, one has*

$$H^i\big(G(L|K), U_L\big) = 1 \quad \text{and} \quad H^i\big(G(L|K), U_L^{(n)}\big) = 1 \quad \text{for} \quad n = 1, 2, \dots$$

In particular,

$$N_{L|K} U_L = U_K \quad \text{and} \quad N_{L|K} U_L^{(n)} = U_K^{(n)} .$$

Proof: Let $G = G(L|K)$. We have already seen that $H^i(G, U_L) = 1$ in chap. IV, (6.2). In order to prove $H^i(G, U_L^{(n)}) = 1$, we first show that

$$H^i(G, \lambda^*) = 1 \quad \text{and} \quad H^i(G, \lambda) = 1,$$

for the residue class field λ of L. It is enough to prove this for $i = -1$, as λ is finite, and so $h(G, \lambda^*) = h(G, \lambda) = 1$. We have $H^{-1}(G, \lambda^*) = 1$ by Hilbert 90 (see chap. IV, (3.5)). Let $f = [\lambda : \kappa]$ be the degree of λ over the residue class field κ of K, and let φ be the Frobenius automorphism of $\lambda|\kappa$. Then we have

$$\#_{N_G}\lambda = \#\Big\{x \in \lambda \;\Big|\; \sum_{i=0}^{f-1} x^{\varphi^i} = \sum_{i=0}^{f-1} x^{q^i} = 0\Big\} \leq q^{f-1}$$

and

$$\#(\varphi - 1)\lambda = q^{f-1},$$

since the map $\lambda \xrightarrow{\varphi - 1} \lambda$ has kernel κ. Therefore $H^{-1}(G, \lambda) = {}_{N_G}\lambda/(\varphi - 1)\lambda = 1$.

Applying now the exact hexagon of chap. IV, (7.1), to the exact sequence of G-modules

$$1 \longrightarrow U_L^{(1)} \longrightarrow U_L \longrightarrow \lambda^* \longrightarrow 1,$$

we obtain $H^i(G, U_L^{(1)}) = H^i(G, U_L) = 1$, because $H^i(G, \lambda^*) = 1$. If π is a prime element of K, then π is also a prime element of L, so the map $U_L^{(n)} \to \lambda$ given by $1 + a\pi^n \mapsto a \bmod \mathfrak{p}_L$ is a G-homomorphism. From the exact sequence

$$1 \longrightarrow U_L^{(n+1)} \longrightarrow U_L^{(n)} \longrightarrow \lambda \longrightarrow 1,$$

we now deduce by induction just as above, because $H^i(G, \lambda) = 0$, that

$$H^i(G, U_L^{(n+1)}) = H^i(G, U_L^{(n)}) = 1,$$

since $H^i(G, U_L^{(1)}) = 1$. □

We now consider the maximal unramified extension $\tilde{k}|k$ over the ground field k. By chap. II, § 9, the residue class field of $\tilde{k}$ is the algebraic closure $\overline{\kappa}$ of the residue class field κ of k. By chap. II, (9.9), we get a canonical isomorphism

$$G(\tilde{k}|k) \cong G(\overline{\kappa}|\kappa) \cong \widehat{\mathbb{Z}}.$$

It associates to the element $1 \in \widehat{\mathbb{Z}}$ the Frobenius automorphism $x \mapsto x^q$ in $G(\overline{\kappa}|\kappa)$, and the Frobenius automorphism φ_k in $G(\tilde{k}|k)$ which is given by

$$a^{\varphi_k} \equiv a^q \bmod \mathfrak{p}_{\tilde{k}}, \quad a \in \mathcal{O}_{\tilde{k}}.$$

For the absolute Galois group $G = G(\bar{k}|k)$ we therefore obtain the continuous and surjective homomorphism

$$d : G \longrightarrow \widehat{\mathbb{Z}}.$$

Thus the abstract notions of chap. IV, § 4, based on this homomorphism, like "unramified", "ramification index", "inertia degree", etc., do agree, in the case at hand, with the corresponding concrete notions defined in chap. II.

As stated above we choose $A = \bar{k}^*$ to be our G-module. Hence $A_K = K^*$, for every finite extension $K|k$. The usual normalized exponential valuation $v_k : k^* \to \mathbb{Z}$ is then henselian with respect to d, in the sense of chap. IV, (4.6). For, given any finite extension $K|k$, $\frac{1}{e_K} v_K$ is the extension of v_k to K^*, and by chap. II, (4.8),

$$\frac{1}{e_K} v_K(K^*) = \frac{1}{[K:k]} v_k(N_{K|k}K^*) = \frac{1}{e_K f_K} v_k(N_{K|k}K^*),$$

i.e., $v_k(N_{K|k}K^*) = f_K v_K(K^*) = f_K\mathbb{Z}$. The pair of homomorphisms

$$(d : G \to \widehat{\mathbb{Z}},\ v_k : k^* \to \mathbb{Z})$$

therefore satisfies all the properties of a class field theory, and we obtain the **Local Reciprocity Law**:

(1.3) Theorem. *For every finite Galois extension $L|K$ of local fields we have a canonical isomorphism*

$$r_{L|K} : G(L|K)^{ab} \xrightarrow{\sim} K^*/N_{L|K}L^*.$$

The general definition of the reciprocity map in chap. IV, (5.6), was actually inspired by the case of local class field theory. This is why it is especially transparent in this case: let $\sigma \in G(L|K)$, and let $\tilde{\sigma}$ be an extension of σ to the maximal unramified extension $\tilde{L}|K$ of L such that $d_K(\tilde{\sigma}) \in \mathbb{N}$

or, in other words, $\tilde{\sigma}|_{\tilde{K}} = \varphi_K^n$, for some $n \in \mathbb{N}$. If Σ is the fixed field of $\tilde{\sigma}$ and $\pi_\Sigma \in \Sigma$ is a prime element, then

$$r_{L|K}(\sigma) = N_{\Sigma|K}(\pi_\Sigma) \bmod N_{L|K}L^* .$$

Inverting $r_{L|K}$ gives us the **local norm residue symbol**

$$(\quad, L|K) : K^* \longrightarrow G(L|K)^{ab} .$$

It is surjective and has kernel $N_{L|K}L^*$.

In global class field theory we will have to take into account the field $\mathbb{R} = \mathbb{Q}_\infty$ along with the p-adic number fields $\mathbb{Q}_p$. It also admits a reciprocity law: for the unique non-trivial Galois extension $\mathbb{C}|\mathbb{R}$, we define the norm residue symbol

$$(\quad, \mathbb{C}|\mathbb{R}) : \mathbb{R}^* \longrightarrow G(\mathbb{C}|\mathbb{R})$$

by

$$(a, \mathbb{C}|\mathbb{R})\sqrt{-1} = \sqrt{-1}^{\,\mathrm{sgn}(a)} .$$

The kernel of $(\ , \mathbb{C}|\mathbb{R})$ is the group $\mathbb{R}_+^*$ of all positive real numbers, which is again the group of norms $N_{\mathbb{C}|\mathbb{R}}\mathbb{C}^* = \{z\bar{z} \mid z \in \mathbb{C}^*\}$.

The reciprocity law gives us a very simple classification of the abelian extensions of a local field K. It is formulated in the following

(1.4) Theorem. *The rule*

$$L \longmapsto \mathcal{N}_L = N_{L|K}L^*$$

gives a $1-1$*-correspondence between the finite abelian extensions of a local field* K *and the open subgroups* $\mathcal{N}$ *of finite index in* K^**. Furthermore,*

$$L_1 \subseteq L_2 \Longleftrightarrow \mathcal{N}_{L_1} \supseteq \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1}\mathcal{N}_{L_2}.$$

Proof: By chap. IV, (6.7), all we have to show is that the subgroups $\mathcal{N}$ of K^* which are open in the norm topology are precisely the subgroups of finite index which are open in the valuation topology. A subgroup $\mathcal{N}$ which is open in the norm topology contains by definition a group of norms $N_{L|K}L^*$. By (1.3), this has finite index in K^*. It is also open because it contains the subgroup $N_{L|K}U_L$ which itself is open, for it is closed, being the image of the compact group U_L, and has finite index in U_K. We prove the converse first in

The case $\mathrm{char}(K) \nmid n$. Let $\mathcal{N}$ be a subgroup of finite index $n = (K^* : \mathcal{N})$. Then $K^{*n} \subseteq \mathcal{N}$, and it is enough to show that K^{*n} contains a group of norms. For this we use Kummer theory (see chap. IV, § 3). We may assume that K^* contains the group μ_n of n-th roots of unity. For if it does not, we put $K_1 = K(\mu_n)$. If K_1^{*n} contains a group of norms $N_{L_1|K_1}L_1^*$, and $L|K$ is a Galois extension containing L_1, then

$$\begin{aligned} N_{L|K}L^* &= N_{K_1|K}(N_{L|K_1}L^*) \subseteq N_{K_1|K}(N_{L_1|K_1}L_1^*) \\ &\subseteq N_{K_1|K}(K_1^{*n}) \subseteq K^{*n} . \end{aligned}$$

So let $\mu_n \subseteq K$, and let $L = K(\sqrt[n]{K^*})$ be the maximal abelian extension of exponent n. Then by chap. IV, § 3, we have

$$(*) \qquad \operatorname{Hom}(G(L|K), \mu_n) \cong K^*/K^{*n} .$$

By chap. II, (5.8), K^*/K^{*n} is finite, and then so is $G(L|K)$. Since $K^*/N_{L|K}L^*$ is isomorphic to $G(L|K)$ and has exponent n, we have that $K^{*n} \subseteq N_{L|K}L^*$, and $(*)$ yields

$$\#K^*/K^{*n} = \#G(L|K) = \#K^*/N_{L|K}L^* ,$$

and therefore $K^{*n} = N_{L|K}L^*$.

The case $\operatorname{char}(K) = p|n$. In this case the proof will follow from Lubin-Tate theory which we will develop in § 4. But it is also possible to do without this theory, at the expense of *ad hoc* arguments which turn out to be somewhat elaborate. Since the result has no further use in the remainder of this book, we simply refer the reader to the beautiful treatment in [122], chap. XI, § 5, and chap. XIV, § 6. □

The proof also shows the following

(1.5) Proposition. *If K contains the n-th roots of unity, and if the characteristic of K does not divide n, then the extension $L = K(\sqrt[n]{K^*})|K$ is finite, and one has*

$$N_{L|K}L^* = K^{*n} \quad \textit{and} \quad G(L|K) \cong K^*/K^{*n} .$$

Theorem (1.4) is called the **existence theorem**, because its essential statement is that, for every open subgroup $\mathcal{N}$ of finite index in K^*, there exists an abelian extension $L|K$ such that $N_{L|K}L^* = \mathcal{N}$. This is the "class field" of $\mathcal{N}$. (Incidentally, when $\operatorname{char}(K) = 0$, every subgroup of finite index is automatically open – see chap. II, (5.7).) Every open subgroup of K^* contains some higher unit group $U_K^{(n)}$, as these form a basis of neighbourhoods of 1 in K^*. We put $U_K^{(0)} = U_K$ and define:

(1.6) Definition. *Let $L|K$ be a finite abelian extension, and n the smallest number ≥ 0 such that $U_K^{(n)} \subseteq N_{L|K}L^*$. Then the ideal*

$$\mathfrak{f} = \mathfrak{p}_K^n$$

is called the **conductor** *of $L|K$.*

(1.7) Proposition. *A finite abelian extension $L|K$ is unramified if and only if its conductor is $\mathfrak{f} = 1$.*

Proof: If $L|K$ is unramified, then $U_K = N_{L|K}U_L$ by (1.2), so that $\mathfrak{f} = 1$. If conversely $\mathfrak{f} = 1$, then $U_K \subseteq N_{L|K}U_L$ and $\pi_K^n \in N_{L|K}L^*$, for $n = (K^* : N_{L|K}L^*)$. If $M|K$ is the unramified extension of degree n, then $N_{M|K}M^* = (\pi_K^n) \times U_K \subseteq N_{L|K}L^*$, and then $M \supseteq L$, i.e., $L|K$ is unramified. □

Every open subgroup $\mathcal{N}$ of finite index in K^* contains a group of the form $(\pi^f) \times U_K^{(n)}$. This is again open and of finite index. Hence every finite abelian extension $L|K$ is contained in the class field of such a group $(\pi^f) \times U_K^{(n)}$. Therefore the class fields for the groups $(\pi^f) \times U_K^{(n)}$ are particularly important. We will characterize them explicitly in § 5, as immediate analogues of the cyclotomic fields over $\mathbb{Q}_p$. In the case of the ground field $K = \mathbb{Q}_p$, the class field of the group $(p) \times U_K^{(n)}$ is precisely the field $\mathbb{Q}_p(\mu_{p^n})$ of p^n-th roots of unity:

(1.8) Proposition. *The group of norms of the extension $\mathbb{Q}_p(\mu_{p^n})|\mathbb{Q}_p$ is the group $(p) \times U_{\mathbb{Q}_p}^{(n)}$.*

Proof: Let $K = \mathbb{Q}_p$ and $L = \mathbb{Q}_p(\mu_{p^n})$. By chap. II, (7.13), the extension $L|K$ is totally ramified of degree $p^{n-1}(p-1)$, and if ζ is a primitive p^n-th root of unity, then $1-\zeta$ is a prime element of L of norm $N_{L|K}(1-\zeta) = p$. We now consider the exponential map of $\mathbb{Q}_p$. By chap. II, (5.5), it gives an isomorphism

$$\exp : \mathfrak{p}_K^\nu \longrightarrow U_K^{(\nu)}$$

for $\nu \geq 1$, provided $p \neq 2$, and for $\nu \geq 2$, even if $p = 2$. It transforms the isomorphism $\mathfrak{p}_K^\nu \to \mathfrak{p}_K^{\nu+s-1}$ given by $a \mapsto p^{s-1}(p-1)a$, into the isomorphism $U_K^{(\nu)} \to U_K^{(\nu+s-1)}$ given by $x \mapsto x^{p^{s-1}(p-1)}$, so that $(U_K^{(1)})^{p^{n-1}(p-1)} = U_K^{(n)}$ if $p \neq 2$, and $(U_K^{(2)})^{2^{n-2}} = U_K^{(n)}$ if $p = 2$, $n > 1$

(the case $p = 2$, $n = 1$ is trivial). Consequently, we have $U_K^{(n)} \subseteq N_{L|K}L^*$ if $p \neq 2$. For $p = 2$ we note that

$$U_K^{(2)} = U_K^{(3)} \cup 5U_K^{(3)} = \big(U_K^{(2)}\big)^2 \cup 5\big(U_K^{(2)}\big)^2,$$

because a number that is congruent to 1 mod 4 is congruent to 1 or 5 mod 8. Hence

$$U_K^{(n)} = \big(U_K^{(2)}\big)^{2^{n-1}} \cup 5^{2^{n-2}}\big(U_K^{(2)}\big)^{2^{n-1}}.$$

It is easy to show that $5^{2^{n-2}} = N_{L|K}(2+i)$, so $U_K^{(n)} \subseteq N_{L|K}L^*$ holds also in case $p = 2$. Since $p = N_{L|K}(1-\zeta)$, we have $(p) \times U_K^{(n)} \subseteq N_{L|K}L^*$, and since both groups have index $p^{n-1}(p-1)$ in K^*, we do find that $N_{L|K}L^* = (p) \times U_K^{(n)}$ as claimed. □

As an immediate consequence of this last proposition, we obtain a local version of the famous theorem of *Kronecker-Weber*, to the effect that every finite abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field.

(1.9) Corollary. *Every finite abelian extension of $L|\mathbb{Q}_p$ is contained in a field $\mathbb{Q}_p(\zeta)$, where ζ is a root of unity. In other words:*

The maximal abelian extension $\mathbb{Q}_p^{ab}|\mathbb{Q}_p$ is generated by adjoining all roots of unity.

Proof: For suitable f and n, we have $(p^f) \times U_{\mathbb{Q}_p}^{(n)} \subseteq N_{L|K}L^*$. Therefore L is contained in the class field M of the group

$$(p^f) \times U_{\mathbb{Q}_p}^{(n)} = \big((p^f) \times U_{\mathbb{Q}_p}\big) \cap \big((p) \times U_{\mathbb{Q}_p}^{(n)}\big).$$

By (1.4), M is the composite of the class field for $(p^f) \times U_{\mathbb{Q}_p}$ – this being the unramified extension of degree f – and the class field for $(p) \times U_{\mathbb{Q}_p}^{(n)}$. M is therefore generated by the $(p^f-1)p^n$-th roots of unity. □

From the local Kronecker-Weber theorem, one may readily deduce the global, classical **Theorem of Kronecker-Weber**.

(1.10) Theorem. *Every finite abelian extension $L|\mathbb{Q}$ is contained in a field $\mathbb{Q}(\zeta)$ generated by a root of unity ζ.*

Proof: Let S be the set of all prime numbers p that are ramified in L, and let L_p be the completion of L with respect to some prime lying above p. Then $L_p|\mathbb{Q}_p$ is abelian, and therefore $L_p \subseteq \mathbb{Q}_p(\mu_{n_p})$, for a suitable n_p. Let p^{e_p} be the precise power of p dividing n_p, and let

$$n = \prod_{p\in S} p^{e_p}.$$

We will show that $L \subseteq \mathbb{Q}(\mu_n)$. For this let $M = L(\mu_n)$. Then $M|\mathbb{Q}$ is abelian, and if p is ramified in $M|\mathbb{Q}$, then p must lie in S. If M_p is the completion with respect to a prime of M above p whose restriction to L gives the completion L_p, then

$$M_p = L_p(\mu_n) = \mathbb{Q}_p(\mu_{p^{e_p}n'}) = \mathbb{Q}_p(\mu_{p^{e_p}})\,\mathbb{Q}_p(\mu_{n'}),$$

with $(n', p) = 1$. $\mathbb{Q}_p(\mu_{n'})|\mathbb{Q}_p$ is the maximal unramified subextension of $\mathbb{Q}_p(\mu_{p^{e_p}n'})|\mathbb{Q}_p$. The inertia group I_p of $M_p|\mathbb{Q}_p$ is therefore isomorphic to the group $G(\mathbb{Q}_p(\mu_{p^{e_p}})|\mathbb{Q}_p)$, and consequently has order $\varphi(p^{e_p})$, where φ is Euler's function. Let I be the subgroup of $G(M|\mathbb{Q})$ generated by all I_p, $p \in S$. The fixed field of I is then unramified, and hence by Minkowski's theorem from chap. III, (2.18), it equals $\mathbb{Q}$, i.e., $I = G(M|\mathbb{Q})$. On the other hand we have

$$\#I \le \prod_{p\in S} \#I_p = \prod_{p\in S} \varphi(p^{e_p}) = \varphi(n) = \big[\mathbb{Q}(\mu_n) : \mathbb{Q}\big],$$

and therefore $[M : \mathbb{Q}] = [\mathbb{Q}(\mu_n) : \mathbb{Q}]$, so that $M = \mathbb{Q}(\mu_n)$. This shows that $L \subseteq \mathbb{Q}(\mu_n)$. □

The following exercises 1–3 presuppose exercises 4–8 of chap. IV, § 3.

Exercise 1. For the Galois group $\Gamma = G(\widetilde{K}|K)$, one has canonically

$$H^1(\Gamma, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z} \quad \text{and} \quad H^1(\Gamma, \mu_n) \cong U_K K^{*n}/K^{*n},$$

the latter provided that n is not divisible by the residue characteristic.

Exercise 2. For an arbitrary field K and a G_K-module A, put

$$H^1(K, A) = H^1(G_K, A).$$

If K is a $\mathfrak{p}$-adic number field and n a natural number, then there exists a nondegenerate pairing

$$H^1(K, \mathbb{Z}/n\mathbb{Z}) \times H^1(K, \mu_n) \longrightarrow \mathbb{Z}/n\mathbb{Z}$$

of finite groups given by

$$(\chi, a) \mapsto \chi((a, \widetilde{K}|K)).$$

If n is not divisible by the residue characteristic p, then the orthogonal complement of

$$H^1_{nr}(K, \mathbb{Z}/n) := H^1(G(\widetilde{K}|K), \mathbb{Z}/n\mathbb{Z}) \subseteq H^1(K, \mathbb{Z}/n\mathbb{Z})$$

is the group

$$H^1_{nr}(K, \mu_n) := H^1(G(\widetilde{K}|K), \mu_n) \subseteq H^1(K, \mu_n).$$

Exercise 3. If $L|K$ is a finite extension of $\mathfrak{p}$-adic number fields, then one has a commutative diagram

$$\begin{array}{ccccc} H^1(L,\mathbb{Z}/n\mathbb{Z}) & \times & H^1(L,\mu_n) & \to & \mathbb{Z}/n\mathbb{Z} \\ \uparrow & & \downarrow N_{L|K} & & \| \\ H^1(K,\mathbb{Z}/n\mathbb{Z}) & \times & H^1(K,\mu_n) & \to & \mathbb{Z}/n\mathbb{Z}. \end{array}$$

Exercise 4 (Local Tate Duality). Show that the statements of exercises 2 and 3 generalize to an arbitrary finite G_K-module A instead of $\mathbb{Z}/n\mathbb{Z}$, and $A' = \mathrm{Hom}(A, \bar{K}^*)$ instead of μ_n.

Hint: Use exercises 4–8 of chap. IV, §3.

Exercise 5. Let $L|K$ be the composite of all $\mathbb{Z}_p$-extensions of a $\mathfrak{p}$-adic number field K (i.e., extensions with Galois group isomorphic to $\mathbb{Z}_p$). Show that the Galois group $G(L|K)$ is a free, finitely generated $\mathbb{Z}_p$-module and determine its rank.

Hint: Use chap. II, (5.7).

Exercise 6. There is only one unramified $\mathbb{Z}_p$-extension of K. Generate it by roots of unity.

Exercise 7. Let p be the residue characteristic of K, and let L be the field generated by all roots of unity of p-power order. The fixed field of the torsion subgroup of $G(L|K)$ is a $\mathbb{Z}_p$-extension. It is called the **cyclotomic** $\mathbb{Z}_p$-extension.

Exercise 8. Let $\widehat{\mathbb{Q}}_p|\mathbb{Q}_p$ be the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}_p$, let $G(\widehat{\mathbb{Q}}_p|\mathbb{Q}_p) \cong \mathbb{Z}_p$ be a chosen isomorphism, and let $\hat{d} : G_{\mathbb{Q}_p} \to \mathbb{Z}_p$ be the induced homomorphism of the absolute Galois group. Show:

For a suitable topological generator u of the group of principal units of $\mathbb{Q}_p$,

$$\hat{v} : \mathbb{Q}_p^* \to \mathbb{Z}_p, \quad \hat{v}(a) = \frac{\log a}{\log u},$$

defines a henselian valuation with respect to $\hat{d}$, in the sense of abstract p-class field theory (see chap. IV, §5, exercise 2).

Exercise 9. Determine all p-class field theories $(d : G_K \to \mathbb{Z}_p,\ v : K^* \to \mathbb{Z}_p)$ over a p-adic number field K.

Exercise 10. Determine all class field theories $(d : G_K \to \widehat{\mathbb{Z}},\ v : K^* \to \widehat{\mathbb{Z}})$ over a $\mathfrak{p}$-adic number field K.

Exercise 11. The **Weil group** of a local field K is the preimage W_K of $\mathbb{Z}$ under the mapping $d_K : G_K \to \widehat{\mathbb{Z}}$. Show:

The norm residue symbol $(\ ,K^{ab}|K)$ of the maximal abelian extension $K^{ab}|K$ yields an isomorphism

$$(\ ,K^{ab}|K) : K^* \xrightarrow{\sim} W_K^{ab},$$

which maps the unit group U_K onto the inertia group $I(K^{ab}|K)$, and the group of principal units $U_K^{(1)}$ onto the ramification group $R(K^{ab}|K)$.

§2. The Norm Residue Symbol over $\mathbb{Q}_p$

If ζ is a primitive m-th root of unity, with $(m,p)=1$, then $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ is unramified, and the norm residue symbol is obviously given by

$$\big(a, \mathbb{Q}_p(\zeta)|\mathbb{Q}_p\big)\,\zeta = \zeta^{p^{v_p(a)}}.$$

But if ζ is a primitive p^n-th root of unity, then we obtain the norm residue symbol for the extension $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ explicitly in the simple form

$$\big(a, \mathbb{Q}_p(\zeta)|\mathbb{Q}_p\big)\,\zeta = \zeta^{u^{-1}},$$

where $a = up^{v_p(a)}$, and $\zeta^{u^{-1}}$ is the power ζ^r with any rational integer $r \equiv u^{-1} \bmod p^n$. This result is important, not only in the local situation, but it will play an essential rôle when we develop global class field theory (see chap. VI, §5). Unfortunately, there is no direct algebraic proof of this fact known to date. We have to invoke a transcendental method which makes use of the *completion* $\widehat{K}$ of the maximal unramified extension $\widetilde{K}$ of a local field K. We extend the Frobenius $\varphi \in G(\widetilde{K}|K)$ to $\widehat{K}$ by continuity. First we prove the

(2.1) Lemma. *For every $c \in \mathcal{O}_{\widehat{K}}$, resp. every $c \in U_{\widehat{K}}$, the equation*

$$x^{\varphi} - x = c, \quad \text{resp.} \quad x^{\varphi-1} = c,$$

admits a solution in $\mathcal{O}_{\widehat{K}}$, resp. in $U_{\widehat{K}}$. If $x^{\varphi} = x$ for $x \in \mathcal{O}_{\widehat{K}}$, then $x \in \mathcal{O}_K$.

Proof: Let π be a prime element of K. Then π is also a prime element of $\widehat{K}$, and we have the φ-invariant isomorphisms

$$U_{\widehat{K}}/U_{\widehat{K}}^{(1)} \cong \bar{\kappa}^*, \quad U_{\widehat{K}}^{(n)}/U_{\widehat{K}}^{(n+1)} \cong \bar{\kappa}$$

(see chap. II, (3.10)). Let $c \in U_{\widehat{K}}$ and $\bar{c} = c \bmod \mathfrak{p}_{\widehat{K}}$. Since the residue class field $\bar{\kappa}$ of $\widehat{K}$ is algebraically closed, the equation $\bar{x}^{\varphi} = \bar{x}^q = \bar{x} \cdot \bar{c}$ $(q = q_K)$ has a solution $\neq 0$ in $\bar{\kappa} = \mathcal{O}_{\widehat{K}}/\mathfrak{p}_{\widehat{K}}$, i.e.,

$$c = x_1^{\varphi-1} a_1, \quad x_1 \in U_{\widehat{K}}, \quad a_1 \in U_{\widehat{K}}^{(1)}.$$

For similar reasons, we find that $a_1 = x_2^{\varphi-1} a_2$, for some $x_2 \in U_{\widehat{K}}^{(1)}$ and $a_2 \in U_{\widehat{K}}^{(2)}$, so that $c = (x_1 x_2)^{\varphi-1} a_2$. Indeed, putting $a_1 = 1 + b_1\pi$, $x_2 = 1 + y_2\pi$, gives $a_1 x_2^{1-\varphi} \equiv 1 - (y_2^{\varphi} - y_2 - b_1)\pi \bmod \pi^2$, i.e., we have to solve the congruence $y_2^{\varphi} - y_2 - b_1 \equiv 0 \bmod \pi$, or equivalently the equation $\bar{y}_2^q - \bar{y}_2 - \bar{b}_1 = 0$ in $\bar{\kappa}$. This is possible because $\bar{\kappa}$ is algebraically closed. Continuing in this way, we get

$$c = (x_1 x_2 \cdots x_n)^{\varphi - 1} a_n, \quad x_n \in U_{\widetilde{K}}^{(n-1)}, \quad a_n \in U_{\widetilde{K}}^{(n)},$$

and passing to the limit finally gives $c = x^{\varphi-1}$, where $x = \prod_{n=1}^{\infty} x_n \in U_{\widetilde{K}}$. The solvability of the equation $x^\varphi - x = c$ follows analogously, using the isomorphisms $\mathfrak{p}_{\widetilde{K}}^n / \mathfrak{p}_{\widetilde{K}}^{n+1} \cong \bar{\kappa}$.

Now let $x \in \mathcal{O}_{\widehat{K}}$ and $x^\varphi = x$. Then, for every $n \geq 1$, one has

$$(*) \qquad x = x_n + \pi^n y_n \quad \text{with } x_n \in \mathcal{O}_K \text{ and } y_n \in \mathcal{O}_{\widehat{K}}.$$

Indeed, for $n = 1$ we have $x = a + \pi b$, with $a \in \mathcal{O}_{\widetilde{K}}$, $b \in \mathcal{O}_{\widehat{K}}$, and $x^\varphi = x$ implies $a^\varphi \equiv a \bmod \pi$. Hence $a = x_1 + \pi c$, with $x_1 \in \mathcal{O}_K$, $c \in \mathcal{O}_{\widetilde{K}}$, and therefore $x = x_1 + \pi(b+c) = x_1 + \pi y_1$, $y_1 \in \mathcal{O}_{\widehat{K}}$. The equation $x = x_n + \pi^n y_n$ implies furthermore that $y_n^\varphi = y_n$, so that we get as above $y_n = c_n + \pi d_n$, with $c_n \in \mathcal{O}_K$, $d_n \in \mathcal{O}_{\widehat{K}}$, and therefore $x = (x_n + c_n \pi^n) + \pi^{n+1} d_n = x_{n+1} + \pi^{n+1} y_{n+1}$, for some $x_{n+1} \in \mathcal{O}_K$, $y_{n+1} \in \mathcal{O}_{\widehat{K}}$. Now passing to the limit in the equation $(*)$ gives $x = \lim_{n\to\infty} x_n \in \mathcal{O}_K$, because K is complete. $\square$

For a power series $F(X_1, \ldots, X_n) \in \mathcal{O}_{\widehat{K}}[[X_1, \ldots, X_n]]$, let F^φ be the power series in $\mathcal{O}_{\widehat{K}}[[X_1, \ldots, X_n]]$ which arises from F by applying φ to the coefficients of F. A **Lubin-Tate series** for a prime element π of K is by definition a power series $e(X) \in \mathcal{O}_K[[X]]$ with the properties

$$e(X) \equiv \pi X \bmod \deg 2 \quad \text{and} \quad e(X) \equiv X^q \bmod \pi,$$

where $q = q_K$ denotes, as always, the number of elements in the residue class field of K. The totality of all Lubin-Tate series is denoted by $\mathcal{E}_\pi$. In $\mathcal{E}_\pi$ there are in particular the polynomials

$$e(X) = uX^q + \pi(a_{q-1}X^{q-1} + \cdots + a_2 X^2) + \pi X,$$

where $u, a_i \in \mathcal{O}_K$ and $u \equiv 1 \bmod \pi$. These are called the **Lubin-Tate polynomials**. The simplest one among them is the polynomial $X^q + \pi X$. In the case $K = \mathbb{Q}_p$ for example, $e(X) = (1+X)^p - 1$ is a Lubin-Tate polynomial for the prime element p.

(2.2) Proposition. *Let π and $\bar{\pi}$ be prime elements of $\widehat{K}$, and let $e(X) \in \mathcal{E}_\pi$, $\bar{e}(X) \in \mathcal{E}_{\bar{\pi}}$ be Lubin-Tate series. Let $L(X_1, \ldots, X_n) = \sum_{i=1}^n a_i X_i$ be a linear form with coefficients $a_i \in \mathcal{O}_{\widehat{K}}$ such that*

$$\pi L(X_1, \ldots, X_n) = \bar{\pi} L^\varphi(X_1, \ldots, X_n).$$

Then there is a uniquely determined power series $F(X_1, \dots, X_n) \in \mathcal{O}_{\widehat{K}}[[X_1, \dots, X_n]]$ *satisfying*

$$F(X_1, \dots, X_n) \equiv L(X_1, \dots, X_n) \bmod \deg 2,$$

$$e\big(F(X_1, \dots, X_n)\big) = F^{\varphi}\big(\bar{e}(X_1), \dots, \bar{e}(X_n)\big).$$

If the coefficients of $e, \bar{e}, L$ *lie in a complete subring* $\mathcal{O}$ *of* $\mathcal{O}_{\widehat{K}}$ *such that* $\mathcal{O}^{\varphi} = \mathcal{O}$, *then* F *has coefficients in* $\mathcal{O}$ *as well.*

Proof: Let $\mathcal{O}$ be a complete subring of $\mathcal{O}_{\widehat{K}}$ such that $\mathcal{O}^{\varphi} = \mathcal{O}$, which contains the coefficients of $e, \bar{e}, L$. We put $X = (X_1, \dots, X_n)$ and $e(X) = (e(X_1), \dots, e(X_n))$. Let

$$F(X) = \sum_{\nu=1}^{\infty} E_{\nu}(X) \in \mathcal{O}[[X]]$$

be a power series, $E_{\nu}(X)$ its homogeneous part of degree ν, and let

$$F_r(X) = \sum_{\nu=1}^{r} E_{\nu}(X).$$

Clearly, $F(X)$ is a solution of the above problem if and only if $F_1(X) = L(X)$ and

$$(1) \qquad e\big(F_r(X)\big) \equiv F_r^{\varphi}\big(\bar{e}(X)\big) \bmod \deg(r+1)$$

for every $r \geq 1$. We determine the polynomials $E_{\nu}(X)$ inductively. For $\nu = 1$ we are forced to take $E_1(X) = L(X)$. Condition (1) is then satisfied for $r = 1$ by hypothesis. Assume that the $E_{\nu}(X)$, for $\nu = 1, \dots, r$, have already been found, and that they are uniquely determined by condition (1). We then put $F_{r+1}(X) = F_r(X) + E_{r+1}(X)$ with a homogeneous polynomial $E_{r+1}(X) \in \mathcal{O}[X]$ of degree $r+1$ which has yet to be determined. The congruences

$$e\big(F_{r+1}(X)\big) \equiv e\big(F_r(X)\big) + \pi E_{r+1}(X) \bmod \deg(r+2),$$

$$F_{r+1}^{\varphi}\big(\bar{e}(X)\big) \equiv F_r^{\varphi}\big(\bar{e}(X)\big) + \bar{\pi}^{r+1} E_{r+1}^{\varphi}(X) \bmod \deg(r+2)$$

show that $E_{r+1}(X)$ has to satisfy the congruence

$$(2) \qquad G_{r+1}(X) + \pi E_{r+1}(X) - \bar{\pi}^{r+1} E_{r+1}^{\varphi}(X) \equiv 0 \bmod \deg(r+2)$$

with $G_{r+1}(X) = e(F_r(X)) - F_r^{\varphi}(\bar{e}(X)) \in \mathcal{O}[[X]]$. We have $G_{r+1}(X) \equiv 0 \bmod \deg(r+1)$ and

$$(3) \qquad G_{r+1}(X) \equiv F_r(X)^q - F_r^{\varphi}(X^q) \equiv 0 \bmod \pi$$

because $e(X) \equiv \bar{e}(X) \equiv X^q \bmod \pi$ and $\alpha^\varphi \equiv \alpha^q \bmod \pi$ for $\alpha \in \mathcal{O}$. Now let $X^i = X_1^{i_1} \cdots X_n^{i_n}$ be a monomial of degree $r+1$ in $\mathcal{O}[X]$. By (3), the coefficient of X^i in G_{r+1} is of the form $-\pi\beta$, with $\beta \in \mathcal{O}$. Let α be the coefficient of the same monomial X^i in E_{r+1}. Then $\pi\alpha - \bar{\pi}\alpha^\varphi$ is the coefficient of X^i in $\pi E_{r+1} - \bar{\pi} E^\varphi_{r+1}$. Since $G_{r+1}(X) \equiv 0 \bmod \deg(r+1)$, (2) holds if and only if the coefficient α of X^i in E_{r+1} satisfies the equation

$$-\pi\beta + \pi\alpha - \bar{\pi}^{r+1}\alpha^\varphi = 0 \tag{4}$$

for every monomial X^i of degree $r+1$. This equation has a unique solution α in $\mathcal{O}_{\widetilde{K}}$, which actually belongs to $\mathcal{O}$. For if we put $\gamma = \pi^{-1}\bar{\pi}^{r+1}$, we obtain the equation

$$\alpha - \gamma\alpha^\varphi = \beta,$$

which is clearly solved by the series

$$\alpha = \beta + \gamma\beta^\varphi + \gamma^{1+\varphi}\beta^{\varphi^2} + \cdots \in \mathcal{O}$$

(the series converges because $v_{\widetilde{K}}(\gamma) \geq 1$). If α' is another solution, then $\alpha - \alpha' = \gamma(\alpha^\varphi - \alpha'^\varphi)$, hence $v_{\widetilde{K}}(\alpha - \alpha') = v_{\widetilde{K}}(\gamma) + v_{\widetilde{K}}((\alpha - \alpha')^\varphi) = v_{\widetilde{K}}(\gamma) + v_{\widetilde{K}}(\alpha - \alpha')$, i.e., $v_{\widetilde{K}}(\alpha - \alpha') = \infty$ because $v_{\widetilde{K}}(\gamma) \geq 1$, and therefore $\alpha = \alpha'$. As a consequence, for every monomial X^i of degree $r+1$, equation (4) has a unique solution α in $\mathcal{O}$, i.e., there exists a unique $E_{r+1}(X) \in \mathcal{O}[X]$ satisfying (2). This finishes the proof. □

(2.3) Corollary. *Let π and $\bar{\pi}$ be prime elements of K, and let $e \in \mathcal{E}_\pi$, $\bar{e} \in \mathcal{E}_{\bar{\pi}}$ be Lubin-Tate series with coefficients in $\mathcal{O}_K$. Let $\pi = u\bar{\pi}$, $u \in U_K$, and $u = \varepsilon^{\varphi-1}$, $\varepsilon \in U_{\widetilde{K}}$. Then there is a uniquely determined power series $\theta(X) \in \mathcal{O}_{\widetilde{K}}[[X]]$ such that $\theta(X) \equiv \varepsilon X \bmod \deg 2$ and*

$$e \circ \theta = \theta^\varphi \circ \bar{e}.$$

Furthermore, there is a uniquely determined power series $[u](X) \in \mathcal{O}_K[[X]]$ such that $[u](X) \equiv uX \bmod \deg 2$ and

$$\bar{e} \circ [u] = [u] \circ \bar{e}.$$

They satisfy

$$\theta^\varphi = \theta \circ [u].$$

Proof: Putting $L(X) = \varepsilon X$, we have $\pi L(X) = \bar{\pi} L^\varphi(X)$ and the first claim follows immediately from (2.2). In the same way, with the linear form $L(X) = uX$, one obtains the existence and uniqueness of the power series $[u](X) \in \mathcal{O}_K[[X]]$. Finally, defining $\theta_1 = \theta^{\varphi^{-1}} \circ [u]$, we get

$$e \circ \theta_1 = (e \circ \theta)^{\varphi^{-1}} \circ [u] = (\theta^\varphi \circ \bar{e})^{\varphi^{-1}} \circ [u] = (\theta^{\varphi^{-1}} \circ [u])^\varphi \circ \bar{e} = \theta_1^\varphi \circ \bar{e},$$

and thus $\theta_1 = \theta$ because of uniqueness. Hence $\theta^\varphi = \theta \circ [u]$. □

(2.4) Theorem. *Let $a = up^{v_p(a)} \in \mathbb{Q}_p^*$, and let ζ be a primitive p^n-th root of unity. Then one has*

$$\big(a, \mathbb{Q}_p(\zeta)|\mathbb{Q}_p\big)\,\zeta = \zeta^{u^{-1}}.$$

Proof: As $\mathbb{N}$ is dense in $\mathbb{Z}_p$, we may assume that $u \in \mathbb{N}$, $(u,p) = 1$. Let $K = \mathbb{Q}_p$, $L = \mathbb{Q}_p(\zeta)$, and let $\sigma \in G(L|K)$ be the automorphism defined by

$$\zeta^\sigma = \zeta^{u^{-1}}.$$

Since $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ is totally ramified, we have $G(L|K) \cong G(\widetilde{L}|\widetilde{K})$, and we view σ as an element of $G(\widetilde{L}|K)$. Then $\tilde{\sigma} = \sigma\varphi_L \in \mathrm{Frob}(\widetilde{L}|K)$ is an element such that $d_K(\tilde{\sigma}) = 1$ and $\tilde{\sigma}|_L = \sigma$. The fixed field Σ of $\tilde{\sigma}$ is totally ramified because $f_{\Sigma|K} = d_K(\tilde{\sigma}) = 1$ by chap. IV, (4.5). The proof of the theorem is based on the fact that the field Σ can be explicitly generated by a prime element π_Σ which is given by the power series θ of (2.3).

In order to do this, assume $\tilde{\sigma}$ and $\varphi = \varphi_L$ have been extended continuously to the completion $\widehat{L}$ of $\widetilde{L}$, and consider the two Lubin-Tate polynomials

$$e(X) = upX + X^p \quad \text{and} \quad \bar{e}(X) = (1+X)^p - 1$$

as well as the polynomial $[u](X) = (1+X)^u - 1$. Then $\bar{e}([u](X)) = (1+X)^{up} - 1 = [u](\bar{e}(X))$. By (2.3), there is a power series $\theta(X) \in \mathcal{O}_{\widehat{K}}[[X]]$ such that

$$e \circ \theta = \theta^\varphi \circ \bar{e} \quad \text{and} \quad \theta^\varphi = \theta \circ [u].$$

Substituting the prime element $\lambda = \zeta - 1$ of L, we obtain a prime element of Σ by

$$\pi_\Sigma = \theta(\lambda).$$

Indeed, $[u](\lambda^\sigma) = (1+\lambda^\sigma)^u - 1 = \zeta^{\sigma u} - 1 = \zeta - 1 = \lambda$, and therefore

$$\pi_\Sigma^{\tilde{\sigma}} = \theta^\varphi(\lambda^\sigma) = \theta\big([u](\lambda^\sigma)\big) = \theta(\lambda) = \pi_\Sigma,$$

i.e., $\pi_\Sigma \in \Sigma$. We will show that

$$P(X) = e^{n-1}(X)^{p-1} + up \in \mathbb{Z}_p[X]$$

is the minimal polynomial of π_Σ, where $e^i(X)$ is defined by $e^0(X) = X$ and $e^i(X) = e(e^{i-1}(X))$. $P(X)$ is monic of degree $p^{n-1}(p-1)$ and irreducible by Eisenstein's criterion, as $e(X) \equiv X^p \bmod p$, and so $e^{n-1}(X)^{p-1} \equiv X^{p^{n-1}(p-1)} \bmod p$. Finally, $e^n(X) = e^{n-1}(X)\cdot(up + e^{n-1}(X)^{p-1}) = e^{n-1}(X)P(X)$, so that

$$P(\pi_\Sigma)e^{n-1}(\pi_\Sigma) = e^n(\pi_\Sigma).$$

Since $e^i(\pi_\Sigma) = e^i(\theta(\lambda)) = \theta^{\varphi^i}(\bar{e}^i(\lambda)) = \theta^{\varphi^i}((1+\lambda)^{p^i} - 1) = \theta^{\varphi^i}(\zeta^{p^i} - 1)$, we have $e^n(\pi_\Sigma) = 0$, $e^{n-1}(\pi_\Sigma) \neq 0$, and thus $P(\pi_\Sigma) = 0$.

Observing that $N_{L|K}(\zeta-1)=(-1)^d p$, $d=[L:K]$ (see chap. II, (7.13)), we obtain

$$N_{\Sigma|K}(\pi_\Sigma)=(-1)^d P(0)=(-1)^d pu \equiv u \bmod N_{L|K}L^*$$

and therefore $r_{L|K}(\sigma)=u \bmod N_{L|K}L^*$, i.e., $(u,L|K)=(a,L|K)=\sigma$, as required. □

In order to really understand this proof of theorem (2.4), one has to read § 4. Let us note that one would get a direct, purely algebraic proof, if one could show without using the power series θ that the splitting field of the polynomial $e^n(X)$ is abelian, and that its elements are all fixed under $\tilde{\sigma}=\sigma\varphi_L$. This splitting field would then have to be equal to the field Σ and every zero of $P(X)=e^n(X)/e^{n-1}(X)$ would have to be a prime element $\pi_\Sigma\in\Sigma$ such that $N_{\Sigma|K}(\pi_\Sigma)\equiv u \bmod N_{L|K}L^*$, in which case $r_{L|K}(\sigma)\equiv u \bmod N_{L|K}L^*$, and so $(u,L|K)=\sigma$.

Exercise 1. The p-class field theory $(d:G_{\mathbb{Q}_p}\to\mathbb{Z}_p,\, v:\mathbb{Q}_p^*\to\mathbb{Z})$ for the unramified $\mathbb{Z}_p$-extension of $\mathbb{Q}_p$, and the p-class field theory $(\hat{d}:G_{\mathbb{Q}_p}\to\mathbb{Z}_p,\, \hat{v}:\mathbb{Q}_p^*\to\mathbb{Z}_p)$ for the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}_p$ (see § 1, exercise 7) yield the same norm residue symbol $(\ ,L|K)$.

Hint: Show that this statement is equivalent to formula (2.4): $(u,\mathbb{Q}_p(\zeta)|\mathbb{Q}_p)\zeta=\zeta^{u^{-1}}$.

Exercise 2. Let $L|K$ be a totally ramified Galois extension, and let $\widehat{L}$ (resp. $\widehat{K}$) be the completion of the maximal unramified extension $\tilde{L}$ (resp. $\tilde{K}$) of L (resp. K). Show that $N_{\widehat{L}|\widehat{K}}\widehat{L}^*=\widehat{K}^*$, and that every $y\in\widehat{L}^*$ with $N_{\widehat{L}|\widehat{K}}(y)=1$ is of the form $y=\prod_i z_i^{\sigma_i-1}$, $\sigma_i\in G(L|K)$.

Exercise 3 (Theorem of Dwork). Let $L|K$ be a totally ramified abelian extension of $\mathfrak{p}$-adic number fields. Let $x\in K^*$ and $y\in\widehat{L}^*$ such that $N_{\widehat{L}|\widehat{K}}(y)=x$. Let $z_i\in\widehat{L}^*$ and choose $\sigma_i\in G(L|K)$ such that

$$y^{\varphi_K-1}=\prod_i z_i^{\sigma_i-1}.$$

Putting $\sigma=\prod_i\sigma_i$, one has $(x,L|K)=\sigma^{-1}$.

Hint: See chap. IV, § 5, exercise 1.

Exercise 4. Deduce from exercises 2 and 3 the formula $(u,\mathbb{Q}_p(\zeta)|\mathbb{Q}_p)\zeta=\zeta^{u^{-1}}$, for some p^n-th root of unity ζ.

§3. The Hilbert Symbol

Let K be a local field, or $K = \mathbb{R}$, $K = \mathbb{C}$. We assume that K contains the group μ_n of n-th roots of unity, where n is a natural number which is relatively prime to the characteristic of K (i.e., n can be arbitrary if $\mathrm{char}(K) = 0$). Over such a field K we then have at our disposal, on the one hand, Kummer theory (see chap. IV, §3), and on the other, class field theory. It is the interplay between both theories, which gives rise to the "Hilbert symbol". This is a highly remarkable phenomenon which will lead us to a generalization of the classical reciprocity law of Gauss, to n-th power residues.

Let $L = K(\sqrt[n]{K^*})$ be the maximal abelian extension of exponent n. By (1.5), we then have

$$N_{L|K}L^* = K^{*n},$$

and class field theory gives us the canonical isomorphism

$$G(L|K) \cong K^*/K^{*n}.$$

On the other hand, Kummer theory gives the canonical isomorphism

$$\mathrm{Hom}(G(L|K), \mu_n) \cong K^*/K^{*n}.$$

The bilinear map

$$G(L|K) \times \mathrm{Hom}(G(L|K), \mu_n) \longrightarrow \mu_n, \quad (\sigma, \chi) \longmapsto \chi(\sigma),$$

therefore defines a nondegenerate bilinear pairing

$$\left(\frac{\ ,\ }{\mathfrak{p}}\right) : K^*/K^{*n} \times K^*/K^{*n} \longrightarrow \mu_n$$

(bilinear in the multiplicative sense). This pairing is called the **Hilbert symbol.** Its relation to the norm residue symbol is described explicitly in the following proposition.

(3.1) Proposition. *For $a, b \in K^*$, the Hilbert symbol $\left(\frac{a,b}{\mathfrak{p}}\right) \in \mu_n$ is given by*

$$\big(a, K(\sqrt[n]{b})|K\big)\sqrt[n]{b} = \left(\frac{a,b}{\mathfrak{p}}\right)\sqrt[n]{b}.$$

Proof: The image of a under the isomorphism $K^*/K^{*n} \cong G(L|K)$ of class field theory is the norm residue symbol $\sigma = (a, L|K)$. The image of b under the isomorphism $K^*/K^{*n} \cong \mathrm{Hom}(G(L|K), \mu_n)$ of Kummer theory is the character $\chi_b : G(L|K) \to \mu_n$ given by $\chi_b(\tau) = \tau\sqrt[n]{b}/\sqrt[n]{b}$. By definition of the Hilbert symbol, we have

$$\left(\frac{a,b}{\mathfrak{p}}\right) = \chi_b(\sigma) = \sigma\sqrt[n]{b}\,/\sqrt[n]{b},$$

hence $(a, K(\sqrt[n]{b})|K)\sqrt[n]{b} = (a, L|K)\sqrt[n]{b} = \left(\frac{a,b}{\mathfrak{p}}\right)\sqrt[n]{b}$. □

The Hilbert symbol has the following fundamental properties:

(3.2) Proposition.

(i) $\left(\frac{aa',b}{\mathfrak{p}}\right) = \left(\frac{a,b}{\mathfrak{p}}\right)\left(\frac{a',b}{\mathfrak{p}}\right)$,

(ii) $\left(\frac{a,bb'}{\mathfrak{p}}\right) = \left(\frac{a,b}{\mathfrak{p}}\right)\left(\frac{a,b'}{\mathfrak{p}}\right)$,

(iii) $\left(\frac{a,b}{\mathfrak{p}}\right) = 1 \iff$ *a is a norm from the extension* $K(\sqrt[n]{b})|K$,

(iv) $\left(\frac{a,b}{\mathfrak{p}}\right) = \left(\frac{b,a}{\mathfrak{p}}\right)^{-1}$,

(v) $\left(\frac{a,1-a}{\mathfrak{p}}\right) = 1$ *and* $\left(\frac{a,-a}{\mathfrak{p}}\right) = 1$,

(vi) *If* $\left(\frac{a,b}{\mathfrak{p}}\right) = 1$ *for all* $b \in K^*$, *then* $a \in K^{*n}$.

Proof: (i) and (ii) are clear from the definition, (iii) follows from (3.1), and (vi) reformulates the nondegenerateness of the Hilbert symbol.

If $b \in K^*$ and $x \in K$ such that $x^n - b \neq 0$, then

$$x^n - b = \prod_{i=0}^{n-1}(x - \zeta^i\beta), \quad \beta^n = b,$$

for some primitive n-th root of unity ζ. Let d be the greatest divisor of n such that $y^d = b$ has a solution in K, and let $n = dm$. Then the extension $K(\beta)|K$ is cyclic of degree m, and the conjugates of $x - \zeta^i\beta$ are the elements $x - \zeta^j\beta$ such that $j \equiv i \bmod d$. We may therefore write

$$x^n - b = \prod_{i=0}^{d-1} N_{K(\beta)|K}(x - \zeta^i\beta).$$

Hence $x^n - b$ is a norm from $K(\sqrt[n]{b})|K$, i.e.,

$$\left(\frac{x^n - b, b}{\mathfrak{p}}\right) = 1.$$

Choosing $x = 1$, $b = 1 - a$, and $x = 0$, $b = -a$ then yield (v). (iv) finally follows from

$$\left(\frac{a,b}{\mathfrak{p}}\right)\left(\frac{b,a}{\mathfrak{p}}\right) = \left(\frac{a,-a}{\mathfrak{p}}\right)\left(\frac{a,b}{\mathfrak{p}}\right)\left(\frac{b,a}{\mathfrak{p}}\right)\left(\frac{b,-b}{\mathfrak{p}}\right)$$
$$= \left(\frac{a,-ab}{\mathfrak{p}}\right)\left(\frac{b,-ab}{\mathfrak{p}}\right) = \left(\frac{ab,-ab}{\mathfrak{p}}\right) = 1. \qquad \square$$

In the case $K = \mathbb{R}$ we have $n = 1$ or $n = 2$. For $n = 1$ one finds, of course, $\left(\frac{a,b}{\mathfrak{p}}\right) = 1$, and for $n = 2$ we have

$$\left(\frac{a,b}{\mathfrak{p}}\right) = (-1)^{\frac{\operatorname{sgn} a-1}{2} \cdot \frac{\operatorname{sgn} b-1}{2}},$$

because $(a, \mathbb{R}(\sqrt{b})|\mathbb{R}) = 1$ for $b > 0$, and $= (-1)^{\frac{\operatorname{sgn} a-1}{2}}$ for $b < 0$. Here the letter $\mathfrak{p}$ symbolically stands for an infinite place.

Next we determine the Hilbert symbol explicitly in the case where K is a local field ($\neq \mathbb{R}, \mathbb{C}$) whose residue characteristic p does not divide n. We call this the case of the **tame Hilbert symbol**. Since $\mu_n \subseteq \mu_{q-1}$ one has $n \,|\, q - 1$ in that case. First we establish the

(3.3) Lemma. *Let $(n, p) = 1$ and $x \in K^*$. The extension $K(\sqrt[n]{x})|K$ is unramified if and only if $x \in U_K K^{*n}$.*

Proof: Let $x = uy^n$ with $u \in U_K$, $y \in K^*$, so that $K(\sqrt[n]{x}) = K(\sqrt[n]{u})$. Let κ' be the splitting field of the polynomial $X^n - u \bmod \mathfrak{p}$ over the residue class field κ, and let $K'|K$ be the unramified extension with residue class field κ' (see chap. II, §9, p. 173). By Hensel's lemma, $X^n - u$ splits over K' into linear factors, so $K(\sqrt[n]{u}) \subseteq K'$ is unramified. Assume conversely that $L = K(\sqrt[n]{x})$ is unramified over K, and let $x = u\pi^r$, where $u \in U_K$ and π is a prime element of K. Then $v_L(\sqrt[n]{u\pi^r}) = \frac{1}{n} v_L(\pi^r) = \frac{r}{n} \in \mathbb{Z}$, hence $n|r$, i.e., $\pi^r \in K^{*n}$, and thus $x \in U_K K^{*n}$. □

Since $U_K = \mu_{q-1} \times U_K^{(1)}$, every unit $u \in U_K$ has a unique decomposition

$$u = \omega(u)\langle u \rangle$$

with $\omega(u) \in \mu_{q-1}$ and $\langle u \rangle \in U_K^{(1)}$, $u \equiv \omega(u) \bmod \mathfrak{p}$. With this notation we will now prove the

(3.4) Proposition. *If $(n, p) = 1$ and $a, b \in K^*$, then*

$$\left(\frac{a,b}{\mathfrak{p}}\right) = \omega\Big((-1)^{\alpha\beta} \frac{b^\alpha}{a^\beta}\Big)^{(q-1)/n},$$

where $\alpha = v_K(a)$, $\beta = v_K(b)$.

Proof: The function

$$\langle a,b\rangle := \omega\Big((-1)^{\alpha\beta}\frac{b^{\alpha}}{a^{\beta}}\Big)^{(q-1)/n}$$

is obviously bilinear (in the multiplicative sense). We may therefore assume that a and b are prime elements: $a=\pi$, $b=-\pi u$, $u\in U_K$. Since clearly $\langle\pi,-\pi\rangle = \big(\frac{\pi,\,-\pi}{\mathfrak{p}}\big) = 1$, we may restrict to the case $a=\pi$, $b=u$. Let $y=\sqrt[n]{u}$ and $K'=K(y)$. Then we have

$$\langle\pi,u\rangle = \omega(u)^{(q-1)/n} \quad\text{and}\quad (\pi,K'|K)y = \Big(\frac{\pi,u}{\mathfrak{p}}\Big)y.$$

By (3.3), we see that $K'|K$ is unramified and by chap. IV, (5.7), $(\pi,K'|K)$ is the Frobenius automorphism $\varphi=\varphi_{K'|K}$. Consequently,

$$\Big(\frac{\pi,u}{\mathfrak{p}}\Big) = \frac{\varphi y}{y} \equiv y^{q-1} \equiv u^{(q-1)/n} \equiv \omega(u)^{(q-1)/n} \equiv \langle\pi,u\rangle \bmod \mathfrak{p},$$

hence $\big(\frac{\pi,u}{\mathfrak{p}}\big) = \langle\pi,u\rangle$, because μ_{q-1} is mapped isomorphically onto κ^* by $U_K\to\kappa^*$. □

The proposition shows in particular that the Hilbert symbol

$$\Big(\frac{\pi,u}{\mathfrak{p}}\Big) = \omega(u)^{(q-1)/n}$$

(in the case $(n,p)=1$) is independent of the choice of the prime element π. We may therefore put

$$\Big(\frac{u}{\mathfrak{p}}\Big) := \Big(\frac{\pi,u}{\mathfrak{p}}\Big) \quad\text{for}\quad u\in U_K.$$

$\big(\frac{u}{\mathfrak{p}}\big)$ is the root of unity determined by

$$\Big(\frac{u}{\mathfrak{p}}\Big) \equiv u^{(q-1)/n} \bmod \mathfrak{p}_K.$$

We call it the **Legendre symbol**, or the n-th **power residue symbol**. Both names are justified by the

(3.5) Proposition. *Let $(n,p)=1$ and $u\in U_K$. Then one has*

$$\Big(\frac{u}{\mathfrak{p}}\Big) = 1 \iff u \textit{ is an } n\textit{-th power} \bmod \mathfrak{p}_K.$$

Proof: Let ζ be a primitive $(q-1)$-th root of unity, and let $m = \frac{q-1}{n}$. Then ζ^n is a primitive m-th root of unity, and

$$\left(\frac{u}{\mathfrak{p}}\right) = \omega(u)^m = 1 \iff \omega(u) \in \mu_m \iff \omega(u) = (\zeta^n)^i$$
$$\iff u \equiv \omega(u) = (\zeta^i)^n \bmod \mathfrak{p}_K . \qquad \square$$

It is an important, but in general difficult, problem to find explicit formulae for the Hilbert symbol $\left(\frac{a,b}{\mathfrak{p}}\right)$ also in the case $p|n$. Let us look at the case where $n = 2$ and $K = \mathbb{Q}_p$. If $a \in \mathbb{Z}_2$, then $(-1)^a$ means

$$(-1)^a = (-1)^r ,$$

where r is a rational integer $\equiv a \bmod 2$.

(3.6) Theorem. *Let $n = 2$. For $a, b \in \mathbb{Q}_p^*$ we write*

$$a = p^\alpha a', \quad b = p^\beta b', \quad a', b' \in U_{\mathbb{Q}_p} .$$

If $p \neq 2$, then

$$\left(\frac{a,b}{p}\right) = (-1)^{\frac{p-1}{2}\alpha\beta}\left(\frac{a'}{p}\right)^\beta\left(\frac{b'}{p}\right)^\alpha .$$

In particular, one has $\left(\frac{p,p}{p}\right) = (-1)^{(p-1)/2}$ and $\left(\frac{p,u}{p}\right) = \left(\frac{u}{p}\right)$, if u is a unit. If $p = 2$ and $a, b \in U_{\mathbb{Q}_2}$, then

$$\left(\frac{2,a}{2}\right) = (-1)^{(a^2-1)/8} ,$$
$$\left(\frac{a,b}{2}\right) = \left(\frac{b,a}{2}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}} .$$

Proof: The claim for the case $p \neq 2$ is an immediate consequence of (3.4), and will be left to the reader. So let $p = 2$. We put $\eta(a) = \frac{a^2-1}{8}$ and $\varepsilon(a) = \frac{a-1}{2}$. An elementary computation shows that

$$\eta(a_1a_2) \equiv \eta(a_1) + \eta(a_2) \bmod 2 \quad \text{and} \quad \varepsilon(a_1a_2) \equiv \varepsilon(a_1) + \varepsilon(a_2) \bmod 2 .$$

Thus both sides of the equations we have to prove are multiplicative and it is enough to check the claim for a set of generators of $U_{\mathbb{Q}_2}/U_{\mathbb{Q}_2}^2$. $\{5, -1\}$ is such a set. We postpone this for the moment and define $(a,b) = \left(\frac{a,b}{2}\right)$.

We have $(-1,x) = 1$ if and only if x is a norm from $\mathbb{Q}_2(\sqrt{-1})|\mathbb{Q}_2$, i.e., $x = y^2 + z^2$, $y,z \in \mathbb{Q}_2$. Since $5 = 4 + 1$ and $2 = 1 + 1$, we find that $(-1,2) = (-1,5) = 1$. If we had $(-1,-1) = 1$, then it would follow that $(-1,x) = 1$ for all x, i.e., -1 would be a square in $\mathbb{Q}_2^*$, which is not the case. Therefore we have $(-1,-1) = -1$.

We have $(2,2) = (2,-1) = 1$ and $(5,5) = (5,-1) = 1$. It remains therefore to determine $(2,5)$. $(2,5) = 1$ would imply $(2,x) = 1$ for all x, i.e., 2 would be a square in $\mathbb{Q}_2^*$, which is not the case. Hence $(2,5) = -1$.

By direct verification one sees that the values we just found coincide with those of $(-1)^{\eta(a)}$, resp. $(-1)^{\varepsilon(a)\varepsilon(b)}$, in the respective cases.

It remains to show that $U_{\mathbb{Q}_2}/U_{\mathbb{Q}_2}^2$ is generated by $\{5,-1\}$. We set $U = U_{\mathbb{Q}_2}$, $U^{(n)} = U_{\mathbb{Q}_2}^{(n)}$. By chap. II, (5.5), $\exp : 2^n\mathbb{Z}_2 \to U^{(n)}$ is an isomorphism for $n > 1$. Since $a \mapsto 2a$ defines an isomorphism $2^2\mathbb{Z}_2 \to 2^3\mathbb{Z}_2$, $x \mapsto x^2$ defines an isomorphism $U^{(2)} \to U^{(3)}$. It follows that $U^{(3)} \subseteq U^2$. Since $\{1,-1,5,-5\}$ is a system of representatives of $U/U^{(3)}$, U/U^2 is generated by -1 and 5. □

It is much more difficult to determine the n-th Hilbert symbol in the general case. It was discovered only in 1964 by the mathematician *Helmut Brückner*. Since the result has not previously been published in an easily accessible place, we state it here without proof for the case $n = p^\nu$ of odd residue characteristic p of K.

So let $\mu_{p^\nu} \subseteq K$, choose a prime element π of K, and let W be the ring of integers of the maximal unramified subextension T of $K|\mathbb{Q}_p$ (i.e., the ring of Witt vectors over the residue class field of K). Then every element $x \in K$ can be written in the form

$$x = f(\pi),$$

with a Laurent series $f(X) \in W((X))$.

For an arbitrary Laurent series $f(X) = \sum_{i\geq -m} a_i X^i \in W((X))$, let $f^{\mathcal{P}}(X)$ denote the series

$$f^{\mathcal{P}}(X) = \sum_i a_i^{\varphi} X^{ip},$$

where φ is the Frobenius automorphism of W. Further, let $\operatorname{Res}(f\,dX) \in W$ denote the residue of the differential $f\,dX$,

$$d\log f := \frac{f'}{f}\,dX,$$

and

$$\log f := \sum_{i=1}^{\infty} (-1)^{i+1} \frac{(f-1)^i}{i},$$

if $f \in 1 + pW[[X]]$.

Now let ζ be a primitive p^ν-th root of unity. Then $1-\zeta$ is a prime element of $\mathbb{Q}_p(\zeta)$, and thus

$$1-\zeta = \pi^e \varepsilon,$$

for some unit ε of K, where e is the ramification index of $K|\mathbb{Q}_p(\zeta)$. Let $\eta(X) \in W[[X]]$ be a power series such that

$$\varepsilon = \eta(\pi),$$

and let $h(X)$ be the series

$$h(X) = \frac{1}{2}\,\frac{1+(1-X^e\eta(X))^{p^\nu}}{1-(1-X^e\eta(X))^{p^\nu}} = \sum_{i=-\infty}^{\infty} a_i X^i\,, \; a_i \in W\,, \; \lim_{i\to-\infty} a_i = 0\,.$$

With this notation we can now state *Brückner*'s formula for the p^ν-th Hilbert symbol $\left(\frac{x,y}{\mathfrak{p}}\right)$, $p = \operatorname{char}(\kappa) \neq 2$.

(3.7) Theorem. *If $x, y \in K^*$ and $f, g \in W((X))^*$ such that $f(\pi) = x$ and $g(\pi) = y$, then*

$$\left(\frac{x,y}{\mathfrak{p}}\right) = \zeta^{w(x,y)}$$

where

$$w(x,y) = Tr_{W|\mathbb{Z}_p} \operatorname{Res} h\cdot\Big(\frac{1}{p}\log\frac{f^p}{f^{\mathcal{P}}}\,d\log g - \frac{1}{p}\log\frac{g^p}{g^{\mathcal{P}}}\,\frac{1}{p}\,d\log f^{\mathcal{P}}\Big) \bmod p^\nu.$$

For the proof of this theorem, we have to refer to [20] (see also [69] and [135]). *Brückner* has also deduced an explicit formula for the case $n = 2^\nu$, but it is much more complicated. A more recent treatment of the theorem, which also includes the case $n = 2^\nu$, has been given by *G. Henniart* [69].

It would be interesting to deduce from these formulae the following classical result of *Iwasawa* [80], *Artin* and *Hasse* (see [9]) relative to the field

$$\Phi_\nu = \mathbb{Q}_p(\zeta),$$

where ζ is a primitive p^ν-th root of unity ($p \neq 2$). Putting $\pi = 1-\zeta$ and denoting by S the trace map from Φ_ν to $\mathbb{Q}_p$, we obtain for the p^ν-th Hilbert symbol $\left(\frac{x,y}{\mathfrak{p}}\right)$ of the field Φ_ν the

(3.8) Proposition. *For $a \in U_{\Phi_\nu}^{(2p^\nu-1)}$ and $b \in \Phi_\nu^*$ one has*

$$\left(\frac{a,b}{\mathfrak{p}}\right) = \zeta^{S(\zeta \log a \, D\log b)/p^\nu}, \tag{1}$$

where $D\log b$ denotes the formal logarithmic derivative in π of an arbitrary representation of b as an integral power series in π with coefficients in $\mathbb{Z}_p$.

For $a \in U_{\Phi_\nu}^{(1)}$ one has furthermore the two **supplementary theorems**

$$\left(\frac{\zeta,a}{\mathfrak{p}}\right) = \zeta^{S(\log a)/p^\nu}, \tag{2}$$

$$\left(\frac{a,\pi}{\mathfrak{p}}\right) = \zeta^{S((\zeta/\pi)\log a)/p^\nu}. \tag{3}$$

The supplementary theorems (2) and (3) go back to *Artin* and *Hasse* [9]. The formula (1) was proved independently by *Artin* [10] and *Hasse* [61] in the case $\nu = 1$, and by *Iwasawa* [80] in general. In the case $\nu = 1$, for instance, one can indeed obtain the formulae from *Brückner*'s theorem (3.7). Since

$$\frac{1}{p} S(\zeta\pi^i) \equiv \begin{cases} 1 \bmod p, & i = p-1, \\ 0 \bmod p, & i \neq p-1, \end{cases} \quad \text{and} \quad \log a \equiv 0 \bmod \mathfrak{p}^2,$$

one may also interpret the ζ-exponent in the formulae (1)–(3) as the $(p-1)$-st coefficient of a π-adic expansion of $\log a \;\, D\log b$. In this way it appears as a formal residue $\operatorname{Res}_\pi \frac{1}{\pi^p} \log a \;\, D\log b$. As to the supplementary theorems, one has to define also $D\log\zeta = -\zeta^{-1}$, $D\log\pi = \pi^{-1}$.

Exercise 1. For $n = 2$ the Hilbert symbol has the following concrete meaning:

$$\left(\frac{a,b}{\mathfrak{p}}\right) = 1 \iff ax^2 + by^2 - z^2 = 0 \quad \text{has a nontrivial solution in } K.$$

Exercise 2. Deduce proposition (3.8) from theorem (3.7).

Exercise 3. Let K be a local field of characteristic p, let $\bar{K}$ be its separable closure, and let $W_n(\bar{K})$ be the ring of Witt vectors of length n, with the operator $\wp : W_n(\bar{K}) \to W_n(\bar{K})$, $\wp a = Fa - a$ (see chap. IV, §3, exercises 2 and 3). Show that one has $\ker(\wp) = W_n(\mathbb{F}_p)$.

Exercise 4. Abstract Kummer theory (chap. IV, (3.3)) yields for the maximal abelian extension $L|K$ of exponent n a surjective homomorphism

$$W_n(K) \to \operatorname{Hom}(G(L|K), W_n(\mathbb{F}_p)), \quad x \mapsto \chi_x,$$

where one has $\chi_x(\sigma) = \sigma\xi - \xi$ for all $\sigma \in G(L|K)$, with an arbitrary $\xi \in W_n(L)$ such that $\wp\xi = x$. Show that $x \mapsto \chi_x$ has kernel $\wp W_n(K)$.

Exercise 5. Define, for $x \in W_n(K)$ and $a \in K^*$, the symbol $[x,a) \in W_n(\mathbb{F}_p)$ by

$$[x,a) := \chi_x((a, L|K)),$$

where $(\ , L|K)$ is the norm residue symbol. Show:

(i) $[x,a) = (a, K(\xi)|K)\xi - \xi$, if $\xi \in W_n(\overline{K})$ with $\wp\xi = x$.

(ii) $[x+y,a) = [x,a) + [y,a)$.

(iii) $[x,ab) = [x,a) + [x,b)$.

(iv) $[x,a) = 0 \iff a \in N_{K(\xi)|K}K(\xi)^*$, where $\xi \in W_n(\overline{K})$ is an element such that $\wp\xi = x$.

(v) $[x,a) = 0$ for all $a \in K^* \iff x \in \wp W_n(K)$.

(vi) $[x,a) = 0$ for all $x \in W_n(K) \iff a \in K^{*p^n}$.

Exercise 6. Let κ be the residue class field of K and π a prime element such that $K = \kappa((\pi))$. Let

$$d : K \to \Omega^1_{K|\kappa}, \quad f \mapsto df,$$

be the canonical map to the differential module of $K|\kappa$ (see chap. III, §2, p. 200). For every $f \in K$ one has

$$df = f'_\pi d\pi,$$

where f'_π is the formal derivative of f in the expansion according to powers of π with coefficients in κ. Show that for $\omega = (\sum_{i>-\infty} a_i\pi^i)d\pi$, the *residue*

$$\operatorname{Res}\omega := a_{-1}$$

does not depend on the choice of the prime element π.

Exercise 7. Show that in the case $n = 1$ the symbol $[x,a)$ is given by

$$[x,a) = Tr_{\kappa|\mathbb{F}_p}\operatorname{Res}\big(x\frac{da}{a}\big).$$

Remark: Such a formula can also be given for $n \geq 1$ (*P. Kölcze* [88]).

§4. Formal Groups

The most explicit realization of local class field theory we have encountered for the case of cyclotomic fields over the field $\mathbb{Q}_p$, i.e., with the extensions $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$, where ζ is a p^n-th root of unity. The notion of formal group allows us to construct such an explicit cyclotomic theory over an arbitrary local field K by introducing a new kind of roots of unity which are "division points" that do the same for the field K as the p^n-th roots of unity do for the field $\mathbb{Q}_p$.

(4.1) Definiton. *A (1-dimensional, commutative)* **formal group** *over a ring* $\mathcal{O}$ *is a formal power series* $F(X,Y) \in \mathcal{O}[[X,Y]]$ *with the following properties:*

(i) $F(X,Y) \equiv X + Y \bmod \deg 2$,

(ii) $F(X,Y) = F(Y,X)$ *"commutativity"*,

(iii) $F(X, F(Y,Z)) = F(F(X,Y),Z)$ *"associativity"*.

From a formal group one gets an ordinary group by evaluating in a domain where the power series converge. If for instance $\mathcal{O}$ is a complete valuation ring and $\mathfrak{p}$ its maximal ideal, then the operation

$$x \underset{F}{+} y := F(x,y)$$

defines a new structure of abelian group on the set $\mathfrak{p}$.

Examples:

1. $\mathbb{G}_a(X,Y) = X + Y$ (the formal additive group).

2. $\mathbb{G}_m(X,Y) = X + Y + XY$ (the formal multiplicative group). Since

$$X + Y + XY = (1+X)(1+Y) - 1,$$

we have

$$(x \underset{\mathbb{G}_m}{+} y) + 1 = (x+1) \cdot (y+1).$$

So the new operation $\underset{\mathbb{G}_m}{+}$ is obtained from multiplication $\cdot$ via the translation $x \mapsto x + 1$.

3. A power series $f(X) = a_1X + a_2X^2 + \cdots \in \mathcal{O}[[X]]$ whose first coefficient a_1 is a unit admits an "inverse", i.e., there exists a power series

$$f^{-1}(X) = a_1^{-1}X + \cdots \in \mathcal{O}[[X]],$$

such that $f^{-1}(f(X)) = f(f^{-1}(X)) = X$. For every such power series,

$$F(X,Y) = f^{-1}\big(f(X) + f(Y)\big)$$

is a formal group.

(4.2) Definition. *A* **homomorphism** $f : F \to G$ *between two formal groups is a power series* $f(X) = a_1X + a_2X^2 + \cdots \in \mathcal{O}[[X]]$ *such that*

$$f(F(X,Y)) = G\big(f(X), f(Y)\big).$$

In example 3, for instance, the power series f is a homomorphism of the formal group F to the additive group $\mathbb{G}_a$. It is called the *logarithm* of F.

A homomorphism $f : F \to G$ is an *isomorphism* if $a_1 = f'(0)$ is a unit, i.e., if there is a homomorphism $g = f^{-1} : G \to F$ such that

$$f\big(g(X)\big) = g\big(f(X)\big) = X.$$

If the power series $f(X) = a_1X + a_2X^2 + \cdots$ satisfies the equation $f(F(X,Y)) = G(f(X), f(Y))$, but its coefficients belong to an extension ring $\mathcal{O}'$, then we call this a homomorphism *defined over* $\mathcal{O}'$. The following proposition is immediately evident.

(4.3) Proposition. *The homomorphisms $f : F \to F$ of a formal group F over $\mathcal{O}$ form a ring* $\mathrm{End}_{\mathcal{O}}(F)$ *in which addition and multiplication are defined by*

$$(f \underset{F}{+} g)(X) = F\big(f(X), g(X)\big), \quad (f \circ g)(X) = f\big(g(X)\big).$$

(4.4) Definition. *A* **formal** $\mathcal{O}$**-module** *is a formal group F over $\mathcal{O}$ together with a ring homomorphism*

$$\mathcal{O} \longrightarrow \mathrm{End}_{\mathcal{O}}(F), \quad a \longmapsto [a]_F(X),$$

such that $[a]_F(X) \equiv aX \bmod \deg 2$.

A homomorphism (over $\mathcal{O}' \supseteq \mathcal{O}$) *between formal $\mathcal{O}$-modules F, G is a homomorphism $f : F \to G$ of formal groups* (over $\mathcal{O}'$) *in the sense of (4.2) such that*

$$f\big([a]_F(X)\big) = [a]_G\big(f(X)\big) \quad \textit{for all} \quad a \in \mathcal{O}.$$

Now let $\mathcal{O} = \mathcal{O}_K$ be the valuation ring of a local field K, and write $q = (\mathcal{O}_K : \mathfrak{p}_K)$. We consider the following special formal $\mathcal{O}_K$-modules.

(4.5) Definition. *A* **Lubin-Tate module** *over $\mathcal{O}_K$ for the prime element π is a formal $\mathcal{O}_K$-module F such that*

$$[\pi]_F(X) \equiv X^q \bmod \pi.$$

This definition reflects once more the dominating principle of class field theory, to the effect that prime elements correspond to Frobenius elements. In fact, if we reduce the coefficients of some formal $\mathcal{O}$-module F modulo π, we obtain a formal group $\bar{F}(X,Y)$ over the residue class field $\mathbb{F}_q$. The reduction mod π of $[\pi]_F(X)$ is an endomorphism of $\bar{F}$. But on the other hand, $f(X) = X^q$ is clearly an endomorphism of $\bar{F}$, its *Frobenius endomorphism.* Thus F is a Lubin-Tate module if the endomorphism defined by a prime element π gives via reduction the Frobenius endomorphism of $\bar{F}$.

Example: The formal multiplicative group $\mathbb{G}_m$ is a formal $\mathbb{Z}_p$-module with respect to the mapping

$$\mathbb{Z}_p \to \operatorname{End}_{\mathbb{Z}_p}(\mathbb{G}_m), \quad a \mapsto [a]_{\mathbb{G}_m}(X) = (1+X)^a - 1 = \sum_{\nu=1}^{\infty} \tbinom{a}{\nu} X^{\nu}.$$

$\mathbb{G}_m$ is a Lubin-Tate module for the prime element p because

$$[p]_{\mathbb{G}_m}(X) = (1+X)^p - 1 \equiv X^p \bmod p.$$

The following theorem gives a complete and explicit overall view of the totality of all Lubin-Tate modules. Let $e(X), \bar{e}(X) \in \mathcal{O}_K[[X]]$ be Lubin-Tate series for the prime element π of K, and let

$$F_e(X,Y) \in \mathcal{O}_K[[X,Y]] \quad \text{and} \quad [a]_{e,\bar{e}}(X) \in \mathcal{O}_K[[X]]$$

($a \in \mathcal{O}_K$) be the power series (uniquely determined according to (2.2)) such that

$$F_e(X,Y) \equiv X+Y \bmod \deg 2, \qquad e\big(F_e(X,Y)\big) = F_e\big(e(X), e(Y)\big),$$

$$[a]_{e,\bar{e}}(X) \equiv aX \bmod \deg 2, \qquad e\big([a]_{e,\bar{e}}(X)\big) = [a]_{e,\bar{e}}\big(\bar{e}(X)\big).$$

If $e(X) = \bar{e}(X)$ we simply write $[a]_{e,\bar{e}}(X) = [a]_e(X)$.

(4.6) Theorem. (i) *The Lubin-Tate modules for π are precisely the series $F_e(X,Y)$, with the formal $\mathcal{O}_K$-module structure given by*

$$\mathcal{O}_K \longrightarrow \operatorname{End}_{\mathcal{O}_K}(F_e), \quad a \longmapsto [a]_e(X).$$

(ii) *For every $a \in \mathcal{O}_K$ the power series $[a]_{e,\bar{e}}(X)$ is a homomorphism*

$$[a]_{e,\bar{e}} : F_{\bar{e}} \longrightarrow F_e$$

of formal $\mathcal{O}$-modules, and it is an isomorphism if a is a unit.

Proof: If F is any Lubin-Tate module, then $e(X) := [\pi]_F(X) \in \mathcal{E}_\pi$ and $F = F_e$ because $e(F(X,Y)) = F(e(X),e(Y))$, and because of the uniqueness statement of (2.2). For the other claims of the theorem one has to show the following formulae.

(1) $F_e(X,Y) = F_e(Y,X)$,

(2) $F_e(X,F_e(Y,Z)) = F_e(F_e(X,Y),Z)$,

(3) $[a]_{e,\bar{e}}(F_{\bar{e}}(X,Y)) = F_e([a]_{e,\bar{e}}(X),[a]_{e,\bar{e}}(Y))$,

(4) $[a+b]_{e,\bar{e}}(X) = F_e([a]_{e,\bar{e}}(X),[b]_{e,\bar{e}}(X))$,

(5) $[ab]_{e,\bar{\bar{e}}}(X) = [a]_{e,\bar{e}}([b]_{\bar{e},\bar{\bar{e}}}(X))$,

(6) $[\pi]_e(X) = e(X)$.

(1) and (2) show that F_e is a formal group. (3), (4), and (5) show that

$$\mathcal{O}_K \longrightarrow \mathrm{End}_{\mathcal{O}_K}(F_e), \quad a \longmapsto [a]_e,$$

is a homomorphism of rings, i.e., that F_e is a formal $\mathcal{O}_K$-module, and that $[a]_{e,\bar{e}}$ is a homomorphism of formal $\mathcal{O}_K$-modules from $F_{\bar{e}}$ to F_e. Finally, (6) shows that F_e is a Lubin-Tate module.

The proofs of these formulae all follow the same pattern. One checks that both sides of each formula are solutions of the same problem of (2.2), and then deduces their equality from the uniqueness statement. In (6) for instance, both power series commence with the linear form πX and satisfy the condition $e([\pi]_e(X)) = [\pi]_e(e(X))$, resp. $e(e(X)) = e(e(X))$. □

Exercise 1. $\mathrm{End}_{\mathcal{O}}(\mathbb{G}_a)$ consists of all aX such that $a \in \mathcal{O}$.

Exercise 2. Let R be a commutative $\mathbb{Q}$-algebra. Then for every formal group $F(X,Y)$ over R, there exists a unique isomorphism

$$\log_F : F \xrightarrow{\sim} \mathbb{G}_a,$$

such that $\log_F(X) \equiv X \bmod \deg 2$, the **logarithm** of F.

Hint: Let $F_1 = \partial F/\partial Y$. Differentiating $F(F(X,Y),Z) = F(X,F(Y,Z))$ yields $F_1(X,0) \equiv 1 \bmod \deg 1$. Let $\psi(X) = 1 + \sum_{n=1}^{\infty} a_n X^n \in R[[X]]$ be the power series such that $\psi(X)F_1(X,0) = 1$. Then $\log_F(X) = X + \sum_{n=1}^{\infty} \frac{a_n}{n} X^n$ does what we want.

Exercise 3. $\log_{\mathbb{G}_m}(X) = \sum\limits_{n=1}^{\infty} (-1)^{n+1} \frac{X^n}{n} = \log(1+X)$.

Exercise 4. Let π be a prime element of the local field K, and let $f(X) = X + \pi^{-1}X^q + \pi^{-2}X^{q^2} + \cdots$ Then

$$F(X,Y) = f^{-1}(f(X)+f(Y)), \quad [a]_F(X) = f^{-1}(af(X)), \quad a \in \mathcal{O}_K,$$

defines a Lubin-Tate module with logarithm $\log_F = f$.

Exercise 5. Two Lubin-Tate modules over the valuation ring $\mathcal{O}_K$ of a local field K, but for different prime elements π and $\bar{\pi}$, are never isomorphic.

Exercise 6. Two Lubin-Tate modules F_e and $F_{\bar{e}}$ for prime elements π and $\bar{\pi}$ always become isomorphic over $\mathcal{O}_{\widehat{K}}$, where $\widehat{K}$ is the completion of the maximal unramified extension $\widetilde{K}|K$.

Hint: The power series θ of (2.3) yields an isomorphism $\theta : F_{\bar{e}} \to F_e$.

§ 5. Generalized Cyclotomic Theory

Formal groups are relevant for local class field theory in that they allow us to construct a perfect analogue of the theory of the p^n-th cyclotomic field $\mathbb{Q}_p(\zeta)$ over $\mathbb{Q}_p$, with its fundamental isomorphism

$$G\big(\mathbb{Q}_p(\zeta)|\mathbb{Q}_p\big) \xrightarrow{\sim} (\mathbb{Z}/p^n\mathbb{Z})^*$$

(see chap. II (7.13)), replacing $\mathbb{Q}_p$ by an arbitrary local ground field K. The formal groups furnish a generalization of the notion of p^n-th root of unity, and provide an explicit version of the local reciprocity law in the corresponding extensions.

A formal $\mathcal{O}_K$-module gives rise to an ordinary $\mathcal{O}_K$-module if we read the power series over a domain in which they converge. We now choose for this the maximal ideal $\bar{\mathfrak{p}}$ of the valuation ring of the algebraic closure $\bar{K}$ of the given local field K. If $G(X_1, \dots, X_n) \in \mathcal{O}_K[[X_1, \dots, X_n]]$ is a power series with constant coefficient 0, and if $x_1, \dots, x_n \in \bar{\mathfrak{p}}$, then the series $G(x_1, \dots, x_n)$ converges in the complete field $K(x_1, \dots, x_n)$ to an element in $\bar{\mathfrak{p}}$. From the definition of the formal $\mathcal{O}$-modules and their homomorphisms we therefore obtain immediately the

(5.1) Proposition. *Let F be a formal $\mathcal{O}_K$-module. Then the set $\bar{\mathfrak{p}}$ with the operations*

$$x \underset{F}{+} y = F(x, y) \quad \text{and} \quad a \bullet x = [a]_F(x),$$

$x, y \in \bar{\mathfrak{p}}, a \in \mathcal{O}_K$, is an $\mathcal{O}_K$-module in the usual sense. We denote it by $\bar{\mathfrak{p}}_F$.

If $f : F \to G$ is a homomorphism (isomorphism) of formal $\mathcal{O}_K$-modules, then

$$f : \overline{\mathfrak{p}}_F \longrightarrow \overline{\mathfrak{p}}_G, \quad x \longmapsto f(x),$$

is a homomorphism (isomorphism) of ordinary $\mathcal{O}_K$-modules.

The operations in $\overline{\mathfrak{p}}_F$, and particularly scalar multiplication $a \bullet x = [a]_F(x)$, must of course not be confused with the usual operations in the field $\overline{K}$.

We now consider a Lubin-Tate module F for the prime element π of $\mathcal{O}_K$. We define the group of π^n-**division points** by

$$F(n) = \left\{\lambda \in \overline{\mathfrak{p}}_F \;\middle|\; \pi^n \bullet \lambda = 0\right\} = \ker\big([\pi^n]_F\big).$$

This is an $\mathcal{O}_K$-module, and an $\mathcal{O}_K/\pi^n\mathcal{O}_K$-module because it is killed by $\pi^n\mathcal{O}_K$.

(5.2) Proposition. *$F(n)$ is a free $\mathcal{O}_K/\pi^n\mathcal{O}_K$-module of rank* 1.

Proof: An isomorphism $f : F \to G$ of Lubin-Tate modules obviously induces isomorphisms $f : \overline{\mathfrak{p}}_F \to \overline{\mathfrak{p}}_G$ and $f : F(n) \to G(n)$ of $\mathcal{O}_K$-modules. By (4.6), Lubin-Tate modules for the same prime element π are all isomorphic. We may therefore assume that $F = F_e$, with $e(X) = X^q + \pi X = [\pi]_F(X)$. $F(n)$ then consists of the q^n zeroes of the iterated polynomial $e^n(X) = (e \circ \cdots \circ e)(X) = [\pi^n]_F(X)$, which is easily shown, by induction on n, to be separable. Now if $\lambda_n \in F(n) \smallsetminus F(n-1)$, then

$$\mathcal{O}_K \longrightarrow F(n), \quad a \longmapsto a \bullet \lambda_n,$$

is a homomorphism of $\mathcal{O}_K$-modules with kernel $\pi^n\mathcal{O}_K$. It induces a bijective homomorphism $\mathcal{O}_K/\pi^n\mathcal{O}_K \to F(n)$ because both sides are of order q^n. $\square$

(5.3) Corollary. *Associating $a \mapsto [a]_F$ we obtain canonical isomorphisms*

$$\mathcal{O}_K/\pi^n\mathcal{O}_K \longrightarrow \operatorname{End}_{\mathcal{O}_K}\big(F(n)\big) \quad \text{and} \quad U_K/U_K^{(n)} \longrightarrow \operatorname{Aut}_{\mathcal{O}_K}\big(F(n)\big).$$

Proof: The map on the left is an isomorphism since $\mathcal{O}_K/\pi^n\mathcal{O}_K \cong F(n)$ and $\operatorname{End}_{\mathcal{O}_K}(\mathcal{O}_K/\pi^n\mathcal{O}_K) = \mathcal{O}_K/\pi^n\mathcal{O}_K$. The one on the right is obtained by taking the unit groups of these rings. $\square$

Given a Lubin-Tate module F for the prime element π, we now define the **field of π^n-division points** by

$$L_n = K\big(F(n)\big).$$

Since $F(n) \subseteq F(n+1)$ we get a tower of fields

$$K \subseteq L_1 \subseteq L_2 \subseteq \ldots \subseteq L_\pi := \bigcup_{n=1}^{\infty} L_n.$$

These fields are also called the **Lubin-Tate extensions**. They only depend on the prime element π, not on the Lubin-Tate module F. For if G is another Lubin-Tate module for π, then by (4.6), there is an isomorphism $f : F \to G$, $f \in \mathcal{O}_K[[X]]$ such that $G(n) = f(F(n)) \subseteq K(F(n))$, and hence $K(G(n)) = K(F(n))$. If F is the Lubin-Tate module F_e belonging to a Lubin-Tate *polynomial* $e(X) \in \mathcal{E}_\pi$, then $e(X) = [\pi]_F(X)$ and $L_n|K$ is the splitting field of the n-fold iteration

$$e^n(X) = (e \circ \cdots \circ e)(X) = [\pi^n]_F(X).$$

Example: If $\mathcal{O}_K = \mathbb{Z}_p$ and F is the Lubin-Tate module $\mathbb{G}_m$, then

$$e^n(X) = [\,p^n]_{\mathbb{G}_m}(X) = (1+X)^{p^n} - 1.$$

So $\mathbb{G}_m(n)$ consists of the elements $\zeta - 1$, where ζ varies over the p^n-th roots of unity. $L_n|K$ is therefore the p^n-th cyclotomic extension $\mathbb{Q}_p(\mu_{p^n})|\mathbb{Q}_p$. The following theorem shows the complete analogy of Lubin-Tate extensions with cyclotomic fields.

(5.4) Theorem. *$L_n|K$ is a totally ramified abelian extension of degree $q^{n-1}(q-1)$ with Galois group*

$$G(L_n|K) \;\cong\; \mathrm{Aut}_{\mathcal{O}_K}\big(F(n)\big) \;\cong\; U_K/U_K^{(n)},$$

i.e., for every $\sigma \in G(L_n|K)$ there is a unique class u mod $U_K^{(n)}$, with $u \in U_K$ such that

$$\lambda^\sigma = [u]_F(\lambda) \quad \textit{for} \quad \lambda \in F(n).$$

Furthermore the following is true: let F be the Lubin-Tate module F_e associated to the polynomial $e(X) \in \mathcal{E}_\pi$, and let $\lambda_n \in F(n) \smallsetminus F(n-1)$. Then λ_n is a prime element of L_n, i.e., $L_n = K(\lambda_n)$, and

$$\phi_n(X) = \frac{e^n(X)}{e^{n-1}(X)} = X^{q^{n-1}(q-1)} + \cdots + \pi \in \mathcal{O}_K[X]$$

is its minimal polynomial. In particular one has $N_{L_n|K}(-\lambda_n) = \pi$.

Proof: If

$$e(X) = X^q + \pi(a_{q-1}X^{q-1} + \cdots + a_2X^2) + \pi X$$

is a Lubin-Tate polynomial, then

$$\begin{aligned}\phi_n(X) &= \frac{e^n(X)}{e^{n-1}(X)} \\ &= e^{n-1}(X)^{q-1} + \pi\big(a_{q-1}e^{n-1}(X)^{q-2} + \cdots + a_2e^{n-1}(X)\big) + \pi\end{aligned}$$

is an Eisenstein polynomial of degree $q^{n-1}(q-1)$. If F is the Lubin-Tate module associated to e, and $\lambda_n \in F(n) \smallsetminus F(n-1)$, then λ_n is clearly a zero of this Eisenstein polynomial, and is therefore a prime element of the totally ramified extension $K(\lambda_n)|K$ of degree $q^{n-1}(q-1)$. Each $\sigma \in G(L|K)$ induces an automorphism of $F(n)$. We therefore obtain a homomorphism

$$G(L_n|K) \longrightarrow \mathrm{Aut}_{\mathcal{O}_K}\big(F(n)\big) \cong U_K/U_K^{(n)}.$$

It is injective because L_n is generated by $F(n)$, and it is surjective because

$$\#G(L_n|K) \geq \big[K(\lambda_n) : K\big] = q^{n-1}(q-1) = \#U_K/U_K^{(n)}.$$

This proves the theorem. □

Generalizing the explicit norm residue symbol of the cyclotomic fields $\mathbb{Q}_p(\mu_{p^n})|\mathbb{Q}_p$ (see (2.4)), we obtain the following explicit formula for the norm residue symbol of the Lubin-Tate extensions.

(5.5) Theorem. *For the field $L_n|K$ of π^n-division points and for $a = u\pi^{v_K(a)} \in K^*$, $u \in U_K$, one has*

$$(a, L_n|K)\lambda = [u^{-1}]_F(\lambda), \quad \lambda \in F(n).$$

Proof: The proof is the same as that of (2.4). Let $\sigma \in G(L_n|K)$ be the automorphism such that

$$\lambda^\sigma = [u^{-1}]_F(\lambda), \quad \lambda \in F(n).$$

Let $\tilde\sigma$ be an element in $\mathrm{Frob}(\tilde L_n|K)$ such that $\sigma = \tilde\sigma\,|_{L_n}$ and $d_K(\tilde\sigma) = 1$. We view $\tilde\sigma$ as an automorphism of the completion $\widehat{L}_n = L_n\widehat{K}$ of $\tilde L_n$. Let Σ be the fixed field of $\tilde\sigma$. Since $f_{\Sigma|K} = d_K(\tilde\sigma) = 1$, $\Sigma|K$ is totally ramified. It has degree $q^{n-1}(q-1)$ because $\Sigma \cap \tilde K = K$ and $\tilde\Sigma = \Sigma\tilde K = \tilde L_n$. Consequently $[\Sigma : K] = [\tilde L_n : \tilde K] = [L_n : K]$.

Now let $e \in \mathcal{E}_\pi$, $\bar{e} \in \mathcal{E}_{\overline{\pi}}$ be Lubin-Tate series over $\mathcal{O}_K$, where $\pi = u\overline{\pi}$, and let $F = F_{\bar{e}}$. By (2.3), there exists a power series $\theta(X) = \varepsilon X + \cdots \in \mathcal{O}_{\widehat{K}}[[X]]$, with $\varepsilon \in U_{\widehat{K}}$, such that

$$\theta^\varphi = \theta \circ [u]_F \quad \text{and} \quad \theta^\varphi \circ \bar{e} = e \circ \theta \quad (\varphi = \varphi_K).$$

Let $\lambda_n \in F(n) \smallsetminus F(n-1)$. λ_n is a prime element of L_n, and

$$\pi_\Sigma = \theta(\lambda_n)$$

is a prime element of Σ because

$$\pi_\Sigma^{\tilde{\sigma}} = \theta^\varphi(\lambda_n^\sigma) = \theta^\varphi\big([u^{-1}]_F(\lambda_n)\big) = \theta(\lambda_n) = \pi_\Sigma .$$

Since $e^i(\theta(\lambda_n)) = \theta^{\varphi^i}(\bar{e}^i(\lambda_n)) = 0$ for $i = n$, and $\neq 0$ for $i = n-1$, we have $\pi_\Sigma \in F_e(n) \smallsetminus F_e(n-1)$. Hence $\Sigma = K(\pi_\Sigma)$ is the field of π^n-division points of F_e, and $N_{\Sigma|K}(-\pi_\Sigma) = \pi = u\overline{\pi}$ by (5.4). Since $\overline{\pi} = N_{L_n|K}(-\lambda_n) \in N_{L_n|K}L_n^*$, we get

$$r_{L_n|K}(\sigma) = N_{\Sigma|K}(-\pi_\Sigma) = \pi \equiv u \bmod N_{L_n|K}L_n^*,$$

and thus

$$(a, L_n|K) = (\pi^{v_K(a)}, L_n|K)(u, L_n|K) = (u, L_n|K) = \sigma . \qquad \square$$

(5.6) Corollary. *The field $L_n|K$ of π^n-division points is the class field relative to the group $(\pi) \times U_K^{(n)} \subseteq K^*$.*

Proof: For $a = u\pi^{v_K(a)}$ we have

$$a \in N_{L_n|K}L_n^* \iff (a, L_n|K) = 1 \iff [u^{-1}]_F(\lambda) = \lambda \quad \text{for all } \lambda \in F(n)$$
$$\iff [u^{-1}]_F = \mathrm{id}_{F(n)} \iff u^{-1} \in U_K^{(n)} \iff a \in (\pi) \times U_K^{(n)}.$$

$\square$

For the maximal abelian extension $K^{ab}|K$, this gives the following generalization of the local Kronecker-Weber theorem (1.9):

(5.7) Corollary. *The maximal abelian extension of K is the composite*

$$K^{ab} = \widetilde{K}L_\pi ,$$

where L_π is the union $\bigcup_{n=1}^\infty L_n$ of the fields L_n of π^n-division points.

Proof: Let $L|K$ be a finite abelian extension. Then we have $\pi^f \in N_{L|K}L^*$ for suitable f. Since $N_{L|K}L^*$ is open in K^*, and since the $U_K^{(n)}$ form a basis of neighbourhoods of 1, we have $(\pi^f) \times U_K^{(n)} \subseteq N_{L|K}L^*$ for a suitable n. Hence L is contained in the class field of the group $(\pi^f) \times U_K^{(n)} = ((\pi) \times U_K^{(n)}) \cap ((\pi^f) \times U_K)$. The class field of $(\pi) \times U_K^{(n)}$ is L_n, and that of $(\pi^f) \times U_K$ is the unramified extension K_f of degree f. It follows that $L \subseteq K_f L_n \subseteq \widetilde{K} L_\pi = K^{ab}$. □

Exercise 1. Let $F = F_e$ be the Lubin-Tate module for the Lubin-Tate series $e \in \mathcal{E}_\pi$, with the endomorphisms $[a] = [a]_e$. Let $S = o_K[[X]]$ and $S^* = \{g \in S \mid g(0) \in U_K\}$. Show:

(i) If $g \in S$ is a power series such that $g(F(1)) = 0$, then g is divisible by $[\pi]$, i.e., $g(X) = [\pi](X)h(X)$, $h(X) \in S$.

(ii) Let $g \in S$ be a power series such that

$$g(X \underset{F}{+} \lambda) = g(X) \quad \text{for all} \quad \lambda \in F(1),$$

where we write $X \underset{F}{+} \lambda = F(X, \lambda)$. Then there exists a unique power series $h(X)$ in S such that

$$g = h \circ \pi .$$

Exercise 2. If $h(X)$ is a power series in S, then the power series

$$h_1(X) = \prod_{\lambda \in F(1)} h(X \underset{F}{+} \lambda)$$

also belongs to S, and one has $h_1(X \underset{F}{+} \lambda) = h_1(X)$ for all $\lambda \in F(1)$.

Exercise 3. Let $N(h) \in S$ be the power series (uniquely determined by exercise 1 and exercise 2) such that

$$N(h) \circ [\pi] = \prod_{\lambda \in F(1)} h(X \underset{F}{+} \lambda).$$

The mapping $N : S \to S$ is called **Coleman's norm operator**. Show:

(i) $N(h_1 h_2) = N(h_1)N(h_2)$.

(ii) $N(h) \equiv h \bmod \mathfrak{p}$.

(iii) $h \in X^i S^*$ for $i \geq 0 \Rightarrow N(h) \in X^i S^*$.

(iv) $h \equiv 1 \bmod \mathfrak{p}^i$ for $i \geq 1 \Rightarrow N(h) \equiv 1 \bmod \mathfrak{p}^{i+1}$.

(v) For the operators $N^0(h) = h$, $N^n(h) = N(N^{n-1}(h))$, one has

$$N^n(h) \circ [\pi^n] = \prod_{\lambda \in F(n)} h(X \underset{F}{+} \lambda), \quad n \geq 0.$$

(vi) If $h \in X^i S^*$, $i \geq 0$, then $N^{n+1}(h)/N^n(h) \in S^*$ and

$$N^{n+1}(h) \equiv N^n(h) \bmod \mathfrak{p}^{n+1}, \quad n \geq 0.$$

Exercise 4. Let $\lambda \in F(n+1) \smallsetminus F(n)$, $n \geq 0$, and $\lambda_i = [\pi^{n-i}](\lambda) \in F(i+1)$ for $0 \leq i \leq n$. Then λ_i is a prime element of the Lubin-Tate extension $L_{i+1} = K(F(i+1))$, and $\mathcal{O}_{i+1} = \mathcal{O}_K[\lambda_i]$ is the valuation ring of L_{i+1}, with maximal ideal $\mathfrak{p}_{i+1} = \lambda_i \mathcal{O}_{i+1}$. Show:

let $\beta_i \in \pi^{n-i}\mathfrak{p}_1\mathcal{O}_{i+1}$, $0 \leq i \leq n$. Then there exists a power series $h(X) \in S$ such that

$$h(\lambda_i) = \beta_i \quad \text{for} \quad 0 \leq i \leq n.$$

Hint: Write $\beta_i = \pi^{n-i}\lambda_0 h_i(\lambda_i)$, with $h_i(X) \in \mathcal{O}[X]$ and put, for $0 \leq i \leq n$: $g_i(X) = [\pi^{n+1}][\pi^i]/[\pi^{i+1}]$. Then $h = \sum_{i=0}^n h_i g_i$ is a solution.

Exercise 5. Let $\lambda \in F(n+1) \smallsetminus F(n)$ and $\lambda_i = [\pi^{n-i}](\lambda)$, $0 \leq i \leq n$. For every $u \in U_{L_{n+1}}$, there exists a power series $h(X) \in \mathcal{O}[[X]]$ such that

$$N_{n,i}(u) = h(\lambda_i) \quad \text{for} \quad 0 \leq i \leq n,$$

where $N_{n,i}$ is the norm from L_{n+1} to L_{i+1}.

Hint: Write $u = h_1(\lambda)$, $h_1(X) \in \mathcal{O}[X]$, and put $h_2 = N^n(h_1) \in S^*$. Show that $\beta_i = N_{n,i}(u) - h_2(\lambda_i) \in \pi^{n-i}\mathfrak{p}_1\mathcal{O}_{i+1}$. Then by exercise 4 there is a power series $h_3(X) \in \mathcal{O}[[X]]$ such that $\beta_i = h_3(\lambda_i)$, $0 \leq i \leq n$. Show that $h = h_2 + h_3$ works.

Remark: The solutions of these exercises are discussed in detail in [79], 5.2.

§ 6. Higher Ramification Groups

Considering the homomorphism

$$(\quad, L|K) : K^* \longrightarrow G(L|K)$$

defined for an abelian extension $L|K$ of local fields by the norm residue symbol, it is striking that both groups are equipped with a canonical filtration: in the group K^* on the left we have the descending chain

$$(*) \qquad K^* \supseteq U_K = U_K^{(0)} \supseteq U_K^{(1)} \supseteq U_K^{(2)} \supseteq \cdots$$

of *higher unit groups* $U_K^{(i)}$, and on the right there is the descending chain

$$(**) \qquad G(L|K) \supseteq G^0(L|K) \supseteq G^1(L|K) \supseteq G^2(L|K) \supseteq \cdots$$

of *ramification groups* $G^i(L|K)$ in the upper numbering (see chap. II, § 10). The latter arose from the ramification groups in the lower numbering

$$G_i(L|K) = \{\sigma \in G(L|K) \mid v_L(\sigma a - a) \geq i+1 \quad \text{for all} \quad a \in \mathcal{O}_L\}$$

via the strictly increasing function

$$\eta_{L|K}(s) = \int_0^s \frac{dx}{(G_0 : G_x)}$$

by the rule

$$G^i(L|K) = G_{\psi_{L|K}(i)}(L|K),$$

where ψ is the inverse function of η. We will now prove the remarkable arithmetic fact that the norm residue symbol $(\ ,L|K)$ relates both filtrations $(*)$ and $(**)$ in a precise way. To this end we determine (generalizing chap. II, § 10, exercise 1) the higher ramification groups of the Lubin-Tate extensions.

(6.1) Proposition. *Let $L_n|K$ be the field of π^n-division points of a Lubin-Tate module for the prime element π. Then*

$$G_i(L_n|K) = G(L_n|L_k) \quad \text{for} \quad q^{k-1} \le i \le q^k - 1.$$

Proof: By (5.4) and (5.5), the norm residue symbol gives an isomorphism $U_K/U_K^{(k)} \to G(L_k|K)$ for every k. Hence $G(L_n|L_k) = (U_K^{(k)}, L_n|K)$. We therefore have to show that

$$G_i(L_n|K) = \big(U_K^{(k)}, L_n|K\big) \quad \text{for} \quad q^{k-1} \le i \le q^k - 1.$$

Let $\sigma \in G_1(L_n|K)$ and $\sigma = (u^{-1}, L_n|K)$. Then we have necessarily $u \in U_K^{(1)}$ because $(,L_n|K) : U_K/U_K^{(n)} \xrightarrow{\sim} G(L_n|K)$ maps the p-Sylow subgroup $U_K^{(1)}/U_K^{(n)}$ onto the p-Sylow subgroup $G_1(L_n|K)$ of $G(L_n|K)$. Let $u = 1 + \varepsilon\pi^m$, $\varepsilon \in U_K$, and $\lambda \in F(n) \smallsetminus F(n-1)$. Then λ is a prime element of L_n and from (5.4) we get that

$$\lambda^\sigma = [u]_F(\lambda) = F\big(\lambda, [\varepsilon\pi^m]_F(\lambda)\big).$$

If $m \ge n$, then $\sigma = 1$, so that $v_{L_n}(\lambda^\sigma - \lambda) = \infty$. If $m < n$, then $\lambda_{n-m} = [\pi^m]_F(\lambda)$ is a prime element of L_{n-m} and therefore also $[\varepsilon\pi^m]_F(\lambda) = [\varepsilon]_F(\lambda_{n-m})$. As $L_n|L_{n-m}$ is totally ramified of degree q^m we may write $[\varepsilon\pi^m]_F(\lambda) = \varepsilon_0\lambda^{q^m}$ for some $\varepsilon_0 \in U_{L_n}$. Since $F(X,0) = X$, $F(0,Y) = Y$, we have $F(X,Y) = X + Y + XYG(X,Y)$ with $G(X,Y) \in \mathcal{O}_K[[X,Y]]$. Thus

$$\lambda^\sigma - \lambda = F(\lambda, \varepsilon_0\lambda^{q^m}) - \lambda = \varepsilon_0\lambda^{q^m} + a\lambda^{q^m+1}, \quad a \in \mathcal{O}_{L_n},$$

i.e.,

$$i_{L_n|K}(\sigma) := v_{L_n}(\lambda^\sigma - \lambda) = \begin{cases} q^m, & \text{if } m < n, \\ \infty, & \text{if } m \ge n. \end{cases}$$

By chap. II, § 10, we have $G_i(L_n|K) = \{\sigma \in G(L_n|K) \mid i_{L_n|K}(\sigma) \ge i + 1\}$. Now let $q^{k-1} \le i \le q^k - 1$. If $u \in U_K^{(k)}$, then $m \ge k$, i.e., $i_{L_n|K}(\sigma) \ge q^k \ge i + 1$, and so $\sigma \in G_i(L_n|K)$. This proves the inclusion $(U_K^{(k)}, L_n|K) \subseteq G_i(L_n|K)$. If conversely $\sigma \in G_i(L_n|K)$ and $\sigma \ne 1$, then $i_{L_n|K}(\sigma) = q^m > i \ge q^{k-1}$, i.e., $m \ge k$. Consequently $u \in U_K^{(k)}$, and this shows the inclusion $G_i(L_n|K) \subseteq (U_K^{(k)}, L_n|K)$. □

From this proposition we get the following result, which may be considered the main theorem of higher ramification theory.

(6.2) Theorem. *If $L|K$ is a finite abelian extension, then the norm residue symbol*

$$(\quad, L|K) : K^* \longrightarrow G(L|K)$$

maps the group $U_K^{(n)}$ onto the group $G^n(L|K)$, for $n \geq 0$.

Proof: We may assume that $L|K$ is totally ramified. For if $L^0|K$ is the maximal unramified subextension of $L|K$, then we have on the one hand $G^n(L|K) = G^n(L|L^0)$ because $\psi_{L^0|K}(s) = s$ and $\psi_{L|K}(s) = \psi_{L|L^0}(\psi_{L^0|K}(s)) = \psi_{L|L^0}(s)$ (see chap. II, (10.8)). On the other hand, by chap. IV, (6.4), and chap. V, (1.2), we have

$$\left(U_{L^0}^{(n)}, L|L^0\right) = \left(N_{L^0|K} U_{L^0}^{(n)}, L|K\right) = \left(U_K^{(n)}, L|K\right),$$

so we may replace $L|K$ by $L|L^0$.

If now $L|K$ is totally ramified and π_L is a prime element of L, then $\pi = N_{L|K}(\pi_L)$ is a prime element of K and $(\pi) \times U_K^{(m)} \subseteq N_{L|K} L^*$ for m sufficiently big. Therefore $L|K$ is contained in the class field of $(\pi) \times U_K^{(m)}$, which, by (5.6), is equal to the field L_m of π^m-division points of some Lubin-Tate module for π. In view of chap. II, (10.9), and chap. IV, (6.4), we may even assume that $L = L_m$. By (6.1), the norm residue symbol maps the group $U_K^{(n)}$ onto the group

$$G(L_m|L_n) = G_i(L_m|K) \quad \text{for} \quad q^{n-1} \leq i \leq q^n - 1.$$

But we have (see chap. II, § 10)

$$\eta_{L|K}(q^n - 1) = \frac{1}{g_0}(g_1 + \cdots + g_{q^n-1})$$

with $g_i = \#G_i(L|K) = \#G(L_m|L_n) = (q^{m-1} - q^{n-1})(q-1)$ for $q^{n-1} \leq i \leq q^n - 1$. This yields $\eta_{L|K}(q^n - 1) = n$ and thus $(U_K^{(n)}, L|K) = G_{q^n-1}(L|K) = G^n(L|K)$. □

Higher ramification groups $G^t(L|K)$ were introduced for arbitrary real numbers $t \geq -1$. Thus we may ask for which numbers they change. We call these numbers the *jumps* of the filtration $\{G^t(L|K)\}_{t \geq -1}$ of $G(L|K)$. In other words, t is a jump if for all $\varepsilon > 0$, one has

$$G^t(L|K) \neq G^{t+\varepsilon}(L|K).$$

(6.3) Proposition (*Hasse-Arf*). *For a finite abelian extension $L|K$, the jumps of the filtration $\{G^t(L|K)\}_{t\ge -1}$ of $G(L|K)$ are rational integers.*

Proof: As in the proof of (6.2), we may assume (since $G^t(L|K) = G^t(L|L^0)$) that $L|K$ is totally ramified and contained in a Lubin-Tate extension $L_m|K$. If now t is a jump of $\{G^t(L|K)\}$, then by chap. II (10.9), t is also a jump of $\{G^t(L_m|K)\}$. Since by (6.1), the jumps of $\{G_s(L_m|K)\}$ are the numbers $q^n - 1$, for $n = 0, \dots, m-1$ ($q = 2$ is an exception: 0 is not a jump), the jumps of $\{G^t(L_m|K)\}$ are the numbers $\eta_{L_m|K}(q^n - 1) = n$, for $n = 0, \dots, m-1$. □

The theorem of *Hasse-Arf* has an important application to *Artin L-series*, which we will study in chap. VII (see chap. VII, (11.4)).